



Degree Project In Engineering Physics

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Time Delay Calculations In Bimetric Gravity

From the Eikonal Approximation to Wigner Smith Operator

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Abstract

When studying the consistency of gravitational theories that extend beyond Einstein's General Relativity, one pays attention to two important properties: first, the absence of ghosts (states with negative kinetic energy), and second, the respect of causality, i.e. the time delay experienced by an interacting system due to gravity must be positive. Hassan-Rosen bimetric gravity is currently the only known ghost-free theory with two dynamical interacting metrics. While the phenomenological consequences of the theory have been studied extensively, there remains a risk that higher-order interaction terms between the two metrics can introduce a negative time delay and hence break causality.

In this work, we propose to compute the time delay in bimetric gravity. After linearizing the theory to first order around flat background solutions, two propagating modes are identified: the first is massless (identical to that present in General relativity), and the second is massive. Using the gravitational interaction of two massless scalars mediated by the two spin-two modes as a toy model, the time delay is calculated from the eikonal approximation.

After adapting the model to the study of one massless and one massive scalar particle, the gravitational interaction of light with the Sun is modeled, and the time delay is calculated. Using the Cassini space mission constraints, the gravitational frequency shift is calculated, and a relationship between the theory parameters α , the ratio of the Planck masses of the two interacting metrics, and m_{FP} , the FierzPauli mass of the massive mode, is derived to satisfy these constraints.

Finally, in the hope of providing a method to move beyond the eikonal approximation for time delay calculations, we introduce the Wigner-Smith operator. We show that within the eikonal limit, the two particle cut WS operator reproduces the eikonal time delay result. We then move beyond the two-particle-cut to analyze the influence of soft graviton intermediate states.

Sammanfattning

När man studerar konsekvent hos gravitationsteorier som sträcker sig bortom Einsteins allmänna relativitetsteori uppmärksammar man två viktiga egenskaper: dels avsaknaden av så kallade ghosts (tillstånd med negativ kinetisk energi), dels respekten för kausalitet, det vill säga att den tidsfördröjning som ett system upplever på grund av gravitationen är positiv. HassanRosens bimetri-gravitation är för närvarande den enda kända ghost-fria teorin med två dynamiska, interagerande metrikor. Även om teorins fenomenologi har studerats ingående kvarstår risken att högre ordningens interaktionstermer mellan de två metrikorna kan ge upphov till negativ tidsfördröjning och därmed bryta kausaliteten.

I detta arbete föreslår vi en beräkning av tidsfördröjningen i bimetri-gravitation. Efter att teorin linjäriserats till första ordningen kring plana bakgrundslösningar identifieras en massiv och en masslös modus. Genom att använda interaktionen mellan två masslösa skalära fält via gravitationen som en leksaksmodell beräknas tidsfördröjningen med hjälp av eikonale-approximationen. En parametrisk studie genomförs därefter för att undersöka hur teori-parametrarna påverkar tidsfördröjningen.

Efter att modellen anpassats till studiet av en masslös och en massiv skalärpartikel modelleras ljusets gravitationella interaktion med solen, och tidsfördröjningen beräknas. Med hjälp av begränsningen från Cassini-rymdmissionen beräknas den gravitationella frekvensskiftningen, och en relation mellan teori-parametrarna α , förhållandet mellan de två metrikornas Planckmassor, och $m_F P$, FierzPauli-massan hos den massiva modusen härleds för att uppfylla Cassini-kravet.

Slutligen, i hopp om att tillhandahålla en metod för att gå bortom eikonale-approximationen vid beräkningar av tidsfördröjning, introduceras WignerSmith-operatorn (WS). Vi visar att WS-operatorn, inom eikonalegränsen och begränsad till tvåpartikelsnittet, reproducerar den eikonalbaserade tidsfördröjningen. Därefter går vi bortom tvåpartikelsnittet för att analysera påverkan av mjuka gravitoner mellanliggande tillstånd.

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Acronyms

DoF degrees of freedom. 7, 8, 13, 15

GR General Relativity. 7, 8, 13, 17, 33, 37, 42, 64

HRBG HassanRosen Bimetric Gravity. 7–9, 13, 15, 17, 18, 33, 35, 45, 64, 65

QFT Quantum field theory. 7, 8, 55, 62

SM Standard Model. 7

WS WignerSmith. 9, 10, 49, 51–53, 57, 62, 63, 65, 66

Chapter 1

Introduction

1.1 Motivation

Currently, our best description of nature relies on two theories General Relativity (GR), describing the gravitational interaction as the curvature of spacetime, and the Standard Model (SM), describing the elementary particles and their interaction within the framework of Quantum field theory (QFT). Over the years, both theories have been remarkably successful at passing experimental tests[1–5].

However, there is accumulating evidence that this current picture is not yet complete. For instance, its inability to account for observational evidence for dark energy and dark matter, which constitute about 95% of the universe [6–9], indicates the need for new physics.

One possible extension is the modification of the gravitational sector to include another spin-2 field: GR could be seen as a theory of a massless spin-2 field with self-interaction. The addition of another interacting spin-2 field enriches the gravitational sector and could provide answers to the unknown nature of dark energy and dark matter. For a long time such models with two interacting spin-2 fields were dismissed because they were plagued with the BoulwareDeser ghost anomaly. This anomaly refers to the existence of nonphysical propagating degrees of freedom with negative kinetic energy. However, recent progress has overcome this problem: In 2011 Hassan and Rosen provided the first ghost-free bimetric theory of gravity [10].

Following its formulation, the phenomenology of HassanRosen Bimetric Gravity (HRBG) has been extensively investigated with multiple remarkable results. For instance, the special interaction structure of the theory gives rise to 7 propagating degrees of free-

dom (DoF). This is the correct number of DoF to describe a massive spin-2 field (5 DoF) and a massless one (2 DoF). Generally there is no such invariant decomposition. However, around a special class of solutions, the linearized theory factorizes into that of a massive and massless spin-2 fields. The massive mode then constitutes a natural candidate for dark matter. If dark matter is indeed due to this massive mode, then the fact that it interacts predominantly through gravity becomes expected [11]. Furthermore the presence of another spin-2 field translates into the existence of a new metric other than that of GR. This opens the possibility for new matter sector, a “dark sector” that couples to the new metric. Hence, the only possible observational effects of this new “dark sector” will be gravitational, resulting in an alternative for dark matter. Additionally, self-accelerating cosmological solutions for HRBG have been derived providing an explanation for the unknown nature of dark energy. For more details about the cosmological phenomenology of HRBG see [12–17], the PhD thesis [18] and the references within.

1.2 Thesis Aim and Scope

The focus of this thesis will be the calculation of time delay in HRBG. The motivation behind this is twofold. First, since its derivation in 1964 [19] and the subsequent measurement in the solar system [20], gravitational time delay (Shapiro time delay) has provided the strongest constraints on any theory beyond GR. Thus, computing this observable in the context of HRBG provides constraints on theory parameter space. Second, the consistency of a theory is not uniquely judged by the absence of nonphysical DoF. One also needs to investigate the causal structure of the theory i.e. that the new modifications do not result in acausal propagation. In this context the gravitational time delay has been proved to be a remarkable probe. An acausal propagation will result in negative time delay. Multiple modification of standard GR have been put through this test. For instance, in [21], the authors show that higher-order derivative corrections to the standard Hilbert-Einstein action result in negative time delay for high energy scattering process, which can only be mitigated if one considers an infinite tower of massive particles with spin higher than 2. The second point is of high importance because the rich gravitational interaction of the two metrics in HRBG can lead to accusal propagation.

For these reasons, we think that adapting existing techniques for gravitational time delay calculations to HRBG and suggesting new ones is highly important. In this work, inspired by recent progress in the application of QFT techniques to gravitational

scattering, we will compute time delay in the eikonal regime i.e., when the center of mass energy of an interacting system is higher than the momentum exchanged between the individual particles. This regime is useful (i) because it corresponds to high energy interactions where acausal effects are suspected to appear, and (ii) because scattering processes can be characterized by the eikonal phase making calculations and interpretation of the results more straightforward.

The rest of this thesis is structured in the following manner: the second chapter will introduce the relevant theoretical notions from HRBG. Namely, we will introduce the action of the theory, derive the equations of motion, and present the special class of proportional solutions. Conditions for the existence of flat Minkowski background are briefly discussed and then the theory is linearized around proportional flat backgrounds leading to a massive and massless spin-2 fields. Next, we consider a system of two massless scalar particles interacting uniquely through gravity. The Feynman rules are stated for this system, and the gravitational time delay is derived from the eikonal phase.

The third chapter focuses on the adaptation of the theoretical framework developed in the second chapter in the study of the gravitational interaction of a photon with the Sun. The aim is to derive the gravitational frequency shift. This is then compared to the results from the Cassini space mission [20]. A relationship between the theory parameters is derived to ensure the compliance with the experimental restrictions.

The fourth chapter focuses on the novel application of the WignerSmith (WS) operator to time delay application in gravitational settings. The WS operator first introduced in quantum mechanics, provides a new approach to time delay calculations that can go beyond the eikonal approximation. We show that the first particle cut-off ¹ reproduces the eikonal result. We then explore higher particle cut-offs. We conclude this thesis with a discussion of the weaknesses of this work and the potential improvements.

1.3 Notations and Conventions

In this section we present the important definitions and conventions used in the rest of the thesis. Throughout this work, the metric signature used is mostly positive (-+++). Einstein summation convention is also used over Greek letters. We will be mostly interested in the interaction of spin-0 particles via gravity. We are interested

¹What is meant by first particle cut-off will be explained in more details in the fourth chapter. In short the WS operator matrix element is calculated as infinite sum over intermediate states, the first particle cut-off is then the first term of this sum.

in two kinds of gravitons: massive and massless. Massive (massless) gravitons are referred to with $M_{\mu\nu}$ ($G_{\mu\nu}$). When considering an interacting system the momenta of the internal lines are denoted by q . While for external lines, the momenta are denoted by p for spin-0 particles, regardless of whether they are massive or not, l for massive gravitons and k for massless ones. When the nature of the graviton is not specified we use $h_{\mu\nu}$ and r to refer to the particle momenta. The normalization convention used in this work follows the one in [22],

$$2\pi\delta(p'^2 + m'^2)\Theta(p'^0)\langle p'|p\rangle = 2\pi^4\delta^{(4)}(p' - p), \quad (1.1)$$

where Θ is the Heaviside function.

In the first part of this work, we will be interested in scattering process of spin-0 particles interacting via gravity,

$$p_1 + p_2 \rightarrow p_3 + p_4$$

. For the $2 \rightarrow 2$ process, the Mandelstam variables are

$$s = -(p_1 + p_2)^2 = -(p_3 + p_4)^2, \quad (1.2a)$$

$$t = -(p_1 - p_3)^2 = -(p_2 - p_4)^2 = -q^2, \quad (1.2b)$$

$$u = -(p_1 - p_4)^2 = -(p_2 - p_3)^2. \quad (1.2c)$$

q is the momentum transfer $q = p_1 - p_3 = p_4 - p_2$.

In the second part, when we are interested in the application of the WS operator to bimetric gravity in the eikonal regime, we will study inelastic scattering processes of the type

$$p_1 + p_2 \rightarrow p_3 + p_4 + r^{(\lambda)}$$

. λ denotes the polarization of the gravitons. For this process the following conventions are used

$$q_1 = p_1 - p_3, \quad (1.3a)$$

$$q_2 = p_2 - p_4, \quad (1.3b)$$

$$q = \frac{1}{2}(q_1 - q_2). \quad (1.3c)$$

The following definitions are given for the $2 \rightarrow 2$ process, but they are generalized for the inelastic scattering case. For a scattering process between an initial state $|p_1, p_2\rangle$

to a final state $|p_3, p_4\rangle$, the scattering matrix is defined through

$$\langle p_3, p_4 | S | p_1, p_2 \rangle = \langle p_3, p_4 | p_1, p_2 \rangle + i \langle p_3, p_4 | T | p_1, p_2 \rangle, \quad (1.4)$$

where T is the transition matrix. The energy-momentum conservation is factorized from the Transition matrix through

$$\langle p_3, p_4 | T | p_1, p_2 \rangle = 2\pi^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) A(s, t). \quad (1.5)$$

Let O be an operator that conserves energy-momentum between the initial and final states. $\hat{O}$ is defined by

$$2\pi\delta(p_3^2 + m_3^2)\Theta(p_3^0)2\pi\delta(p_4^2 + m_4^2)\Theta(p_4^0) \langle p_3, p_4 | O | p_1, p_2 \rangle = 2\pi^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \hat{O}(s, t). \quad (1.6)$$

Specifically, for the scattering matrix operator we get

$$2\pi\delta(p_3^2 + m_3^2)\Theta(p_3^0)2\pi\delta(p_4^2 + m_4^2)\Theta(p_4^0) \langle p_3, p_4 | S | p_1, p_2 \rangle = 2\pi^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \hat{S}(s, t). \quad (1.7)$$

Using the delta function properties, namely

$$\delta^{(4)}(p_1 + p_2 - p_3 - p_4) \delta^{(4)}(q) = \delta^{(4)}(p_1 - p_3) \delta^{(4)}(p_2 - p_4), \quad (1.8)$$

we get ²

$$\hat{S}(s, t) = 2\pi^4 \delta^{(4)}(q) + 2\pi\delta(p_3^2 + m_3^2)2\pi\delta(p_4^2 + m_4^2)A(s, t), \quad (1.9a)$$

$$\hat{S}(s, -q^2) = 2\pi^4 \delta^{(4)}(q) + 2\pi\delta(2p_1 \cdot q + q^2)2\pi\delta(-2p_2 \cdot q + q^2)A(s, -q^2). \quad (1.9b)$$

For a function $f(s, q)$ we define two types of Fourier transform to impact parameter space

$$\tilde{f}(s, b) = \int \frac{d^4 q}{2\pi^4} e^{ib \cdot q} f(s, q), \quad (1.10)$$

and

$$FT[f](s, b) = \int \frac{d^4 q}{2\pi^4} e^{ib \cdot q} 2\pi\delta(2p_1 \cdot q + q^2)2\pi\delta(-2p_2 \cdot q + q^2) f(s, q), \quad (1.11a)$$

$$\approx \int \frac{d^4 q}{2\pi^4} e^{ib \cdot q} 2\pi\delta(2p_1 \cdot q)2\pi\delta(-2p_2 \cdot q) f(s, q). \quad (1.11b)$$

In going from the first to the second line in the previous equation, we assumed that

² Θ functions have been omitted for simplicity and because in the cases studied in this work the positivity of the energy of particle is taken to be guaranteed given the eikonal approximation.

$p_i \cdot q \gg q^2$, This is valid within the eikonal approximation. This will be discussed in more detail in later sections. Thus we get

$$\tilde{\hat{S}}(s, b) = 1 + FT[A](s, b). \quad (1.12)$$

The generalizations of the previous results to the inelastic process are obtained by taking $\hat{S}$ as a function of the center of mass energy squared s , $q = \frac{1}{2}(q_1 - q_2)$ and the momentum of the outgoing graviton r . The Fourier transforms to the impact parameter space are then performed with respect to q .

Chapter 2

Time Delay Calculations in Bimetric Gravity

The aim of this chapter is to introduce the theoretical framework of HRBG. This first step will mostly be based on the review article [23]. Next, we consider a toy model of two massless scalar particles interacting uniquely through gravity. We derive the eikonal phase of the system and then compute the time delay. This derivation is based on reference [22] and adapted to bimetric gravity.

2.1 Hassan Rosen Bimetric Gravity

2.1.1 Action and Equations of Motions

The HRBG is a theory for gravitational interaction that extends GR by including another dynamical metric. The interaction potential between the two theories is chosen so that the theory gives rise to only physical DoF. The action for HRBG is given by [10]

$$S_{HR} = m_g^2 \int d^4x \sqrt{-g} R(g) + m_f^2 \int d^4x \sqrt{-f} R(f) - 2m^4 \int d^4x \sqrt{-g} \sum_{i=1}^4 \beta_i e_i(\sqrt{g^{-1}f}), \quad (2.1)$$

where

1. $g_{\mu\nu}$ and $f_{\mu\nu}$ are the two metrics representing the two spin-2 fields.
2. m_g and m_f are the Planck masses.

3. $R(g)$ and $R(f)$ are respectively the Ricci scalars for the metrics g and f .
4. m describes the strength of the interaction.
5. g is defined by $g = -\det(g_{\mu\nu})$.
6. f is defined by $f = -\det(f_{\mu\nu})$.
7. $\sqrt{g^{-1}f}$ is matrix defined by $(\sqrt{g^{-1}f})^{2\mu}{}_{\nu} = g^{\mu\rho} f_{\rho\nu}$.

e_n are the elementary symmetric polynomials given by:

$$e_n(S) = \frac{1}{n!(4-n)!} \epsilon^{\mu_1, \dots, \mu_n, \lambda_{n+1}, \dots, \lambda_4} \epsilon_{\nu_1, \dots, \nu_n, \lambda_{n+1}, \dots, \lambda_4} S^{\nu_1}{}_{\mu_1} \dots S^{\nu_n}{}_{\mu_n}. \quad (2.2)$$

A More useful recursive definition of the symmetric elementary polynomials is given by

$$e_0(S) = 1, \quad (2.3a)$$

$$e_n(S) = \frac{(-1)^{n+1}}{n} \sum_{k=0}^{n-1} (-1)^k \text{Tr}(S^{n-k}) e_k(S). \quad (2.3b)$$

Using the recursion formula 2.3 we can prove that the variation of the elementary polynomials is given by

$$\delta e_0(S) = 0, \quad (2.4a)$$

$$\delta e_n(S) = - \sum_{k=1}^n (-1)^k \text{Tr}(S^{k-1} \delta S) e_{n-k}(S). \quad (2.4b)$$

By applying the least action principle, we get the following two equations of motion for the two spin-2 fields

$$R_{\mu\nu}(g) - \frac{1}{2} g_{\mu\nu} R(g) + \frac{m^4}{m_g^2} V_{\mu\nu}^g(g, f, \beta_n) = 0, \quad (2.5a)$$

$$R_{\mu\nu}(f) - \frac{1}{2} f_{\mu\nu} R(f) + \frac{m^4}{m_g^2} V_{\mu\nu}^f(g, f, \beta_n) = 0, \quad (2.5b)$$

with

$$V_{\mu\nu}^g(g, f, \beta_n) = g_{\mu\rho} \sum_{n=0}^3 (-1)^n \beta_n (Y_n)^\rho{}_\nu(S), \quad (2.6a)$$

$$V_{\mu\nu}^f(g, f, \beta_n) = f_{\mu\rho} \sum_{n=0}^3 (-1)^n \beta_{4-n} (Y_n)^\rho{}_\nu(S^{-1}), \quad (2.6b)$$

$$(Y_n)^\rho{}_\nu(S) = \sum_{k=0}^n (-1)^k e_k(S) (S^{n-k})^\rho{}_\nu, \quad (2.6c)$$

$$S = \sqrt{g^{-1}f}. \quad (2.6d)$$

The interaction potentials for the metrics g and f satisfy the following properties [23]:

$$\sqrt{g} g^{\mu\rho} \nabla_\rho V_{\mu\nu}^g = -\sqrt{f} f^{\mu\rho} \bar{\nabla}_\rho V_{\mu\nu}^f, \quad (2.7a)$$

$$V_{\mu\nu}^f(g, f, \beta_n) = V_{\mu\nu}^g(f, g, \beta_{4-n}), \quad (2.7b)$$

where ∇ and $\bar{\nabla}$ are the covariant derivatives compatible with g and f respectively. HRBG corresponds to 7 propagating DoF. This can be seen from a counting of the metric variables and the constraints. Each metric g and f corresponds to 10 DoF due to their symmetric nature. Of these, 4 are constrained by the Bianchi identity 2.7a. Furthermore, due to the interaction potential, the action 2.1 is not invariant under independent coordinate transformations of the 2 metrics but only under those variations that transform the two metric simultaneously. This simultaneous transformation eliminates 8 DoF through gauge transformation and gauge fixing. Thus out of the original 20 DoF we are left with 8. One additional DoF is constrained by the special structure of the interaction potential. Thus in total we are left with 7 DoF, which is the correct number for a theory that describes massive and massless spin-2 particles. For a more detailed account of the ghost free nature of the theory see [23].

2.1.2 Proportional solutions

One of the simplest and most important classes of solutions in bimetric theory is the proportional solutions. This is obtained by assuming that the metrics $f_{\mu\nu}$ and $g_{\mu\nu}$ are conformally related,

$$f_{\mu\nu} = c(x)^2 g_{\mu\nu}, \quad (2.8)$$

where $c(x)$ is a spacetime scalar function. However, given the Bianchi identity

$$\nabla_\mu (R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R) = 0, \quad (2.9)$$

and taking the divergence of the first bimetric equation 2.5a, leads to

$$\nabla^\mu V_{\mu\nu}^g(g, f, \beta_n) = 0. \quad (2.10)$$

Recalling that the interaction potential only depends on $S = \sqrt{g^{-1}f}$ and the proportionality relation between g and f , we deduce that $V_{\mu\nu}^g(g, f, \beta_n)$ depends on space time variables only through the scalar function $c(x)$. Thus, we get

$$\nabla^\mu V_{\mu\nu}^g(g, f, \beta_n) = \nabla^\mu c(x) \frac{dV_{\mu\nu}^g(g, f, \beta_n)}{dc} = 0, \quad (2.11)$$

which leads to

$$\partial^\mu c(x) = 0, \quad \forall \mu \in \{0, 1, 2, 3\}, \quad (2.12)$$

and therefore c is constant. Substituting this result in the dynamical equation for the bimetric theory, we get two copies of Einstein equations with two different cosmological constants:

$$G(g)_{\mu\nu} + \Lambda_g g_{\mu\nu} = 0, \quad (2.13a)$$

$$G(g)_{\mu\nu} + \Lambda_f g_{\mu\nu} = 0. \quad (2.13b)$$

The Einstein tensor G appearing in both equation is the same due to its scale invariant nature. i.e. $G(g)_{\mu\nu} = G(c \cdot g)_{\mu\nu}$. The cosmological constants appearing in the previous equations are expressed through the theory parameters and the constant of proportionality c through:

$$\Lambda_g(c) = \frac{m^4}{m_g^2}(\beta_0 + 3c\beta_1 + 3c^2\beta_2 + c^3\beta_3), \quad (2.14a)$$

$$\Lambda_f(c) = \frac{m^4}{(m_f c)^2}(c\beta_1 + 3c^2\beta_2 + 3c^3\beta_3 + c^4\beta_4). \quad (2.14b)$$

Taking the difference between the 2 dynamical equations 2.13, leads to:

$$\Lambda_g(c) = \Lambda_f(c). \quad (2.15)$$

This final equation 2.15, serves to express the proportionality constant c in terms of the theory parameters β_n and $\alpha = \frac{m_f}{m_g}$.

The importance of the proportional solutions in HRBG is twofold:

1. First, given that these solutions lead to two copies of Einstein equations, they reproduce all of GR solutions. Thus, HRBG is indeed an extension of Einstein's theory.
2. Second, attempts to linearize HRBG are usually blocked by the structure of the interaction potential: Generically the square-root matrix S does not commute with the metrics perturbations $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$. This results in a complex and lengthy procedure. The proportional solutions offer a special case where the linearization is fairly straightforward. This is due to the special nature of the the square-root matrix S that simply equals to the proportionality constant c for proportional solution and hence commutes with all perturbations.

2.1.3 Spectrum of linear mass eigen states

Building on the remarks from the last section, we propose to linearize HRBG around proportional solutions. The calculations in this section are detailed in appendix A.

Consider small perturbations of both metrics $\bar{g}_{\mu\nu}$ and $\bar{f}_{\mu\nu} = c^2 \bar{g}_{\mu\nu}$,

$$g^{\mu\nu} = \bar{g}^{\mu\nu} + \delta g^{\mu\nu}, \quad (2.16a)$$

$$f^{\mu\nu} = c^2 \bar{g}^{\mu\nu} + \delta f^{\mu\nu}. \quad (2.16b)$$

The resulting variation of the square-root matrix S is given by

$$\bar{g}_{\mu\rho}(\delta S)^\rho{}_\nu = \frac{1}{2c} [\delta f_{\mu\nu} - c^2 \delta g_{\mu\nu}]. \quad (2.17)$$

Plugging in these perturbation expansions into the dynamical equations for the bimetric theory and keeping only linear terms in $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$ we get the linearized equation

$$\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta g_{\rho\sigma} - \bar{\Lambda}_g (\delta g_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \bar{G}^{\rho\sigma} \delta g_{\rho\sigma}) - N \bar{G}_{\mu\rho} ((\delta S)^\rho{}_\nu - \delta^\rho{}_\nu (\delta S)_\eta^\eta) = 0, \quad (2.18a)$$

$$\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta f_{\rho\sigma} - \bar{\Lambda}_g (\delta f_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \bar{G}^{\rho\sigma} \delta f_{\rho\sigma}) - \alpha^2 N \bar{G}_{\mu\rho} ((\delta S)^\rho{}_\nu - \delta^\rho{}_\nu (\delta S)_\eta^\eta) = 0. \quad (2.18b)$$

In these equations the background metric has been redefined to

$$\bar{G}_{\mu\nu} = (1 + \alpha^2 c^2) \bar{g}_{\mu\nu}. \quad (2.19)$$

The cosmological constant has also been redefined with respect to this new background,

$$\bar{\Lambda}_g = \frac{1}{1 + \alpha^2 c^2} \Lambda_g. \quad (2.20)$$

The kinetic operator $\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma}$ is defined with the new background as

$$\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} = -\frac{1}{2}[\delta_\mu^\rho \delta_\nu^\sigma \bar{\nabla}^2 + \bar{G}^{\rho\sigma} \bar{\nabla}_\mu \bar{\nabla}_\nu - \delta_\mu^\rho \bar{\nabla}^\sigma \bar{\nabla}_\nu - \delta_\nu^\sigma \bar{\nabla}^\rho \bar{\nabla}_\mu - \bar{G}_{\mu\nu} \bar{G}^{\rho\sigma} \bar{\nabla}^2 + \bar{G}_{\mu\nu} \bar{\nabla}^\rho \bar{\nabla}^\sigma]. \quad (2.21)$$

Where the covariant derivative have been defined with respect to the new background $\bar{G}_{\mu\nu}$. In this linearized form, the bimetric equations can be diagonalized into massive and massless spin-2 field equations. The needed new fields for this diagonalization are given by

$$\delta G_{\mu\nu} = \delta g_{\mu\nu} + \alpha^2 \delta f_{\mu\nu}, \quad (2.22a)$$

$$\delta M_{\mu\nu} = \frac{1}{2c}(\delta f_{\mu\nu} - c^2 \delta g_{\mu\nu}). \quad (2.22b)$$

In these new variables the equations of motions take the simplified form:

$$\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta G_{\rho\sigma} - \bar{\Lambda}_g (\delta G_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \delta G) = 0, \quad (2.23a)$$

$$\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta M_{\rho\sigma} - \bar{\Lambda}_g (\delta M_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \delta M) + \frac{\bar{m}_{FP}^2}{2} (\delta M_{\mu\nu} - \bar{G}_{\mu\nu} \delta M) = 0. \quad (2.23b)$$

The trace of a tensor is defined with respect to the background metric $\bar{G}_{\mu,\nu}$ so $\delta M = \bar{G}_{\mu\nu} \delta M^{\mu\nu}$. The same holds for δG . The Fierz-Pauli mass term is expressed as a function of the theory parameters and N (this also defines N appearing in 2.18),

$$\bar{m}_{FP}^2 = \frac{(1 + \alpha^2 c^2)^2}{\alpha^2 c} N = \frac{m^4}{m_g^2} \frac{1}{\alpha^2 c^2} (c\beta_1 + 2c^2\beta_2 + c^3\beta_3). \quad (2.24)$$

In this expressions it is important to keep in mind that c is not an independent parameter but determined through the background equation 2.15. The linearized HRBG around proportional background solutions does indeed give rise to massive ($\delta M_{\mu\nu}$) and massless ($\delta G_{\mu\nu}$) spin-2 perturbations. Note that these equations are decoupled i.e. there is no interaction between the 2 modes. This is due to the first order expansion in terms of the perturbations, thus for self-interactions and interactions between the 2 modes to appear one needs to go to higher orders.

The linearized action in terms of the mass eigenstates is given by

$$S_{ME} = \frac{1}{2} \int d^4x \, \delta G^{\mu\nu} \left[\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta G_{\rho\sigma} - \bar{\Lambda}_g (\delta G_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \delta G) \right] \\ + \delta M^{\mu\nu} \left[\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma} \delta M_{\rho\sigma} - \bar{\Lambda}_g (\delta M_{\mu\nu} - \frac{1}{2} \bar{G}_{\mu\nu} \delta M) + \frac{\bar{m}_{FP}^2}{2} (\delta M_{\mu\nu} - \bar{G}_{\mu\nu} \delta M) \right]. \quad (2.25)$$

With the linearized theory at hand, we are ready to study the interaction of test particles through gravity

2.1.4 Matter Coupling and Feynman rules

In order to set the stage for the application of the eikonal approximation in the context of linearized bimetric theory, we need to first make sure that flat background solutions exist. It is important because it ensures that the propagator for the massless and massive modes take simple forms from which we can extract the equivalent of Feynman rules. To this end we first start by absorbing the proportionality parameter c in the rest of the theory parameters. Thus we define

$$\beta'_n = c^{-n} \beta_n, \quad (2.26a)$$

$$\alpha' = \alpha c. \quad (2.26b)$$

the ' notation is then dropped and we use the α and β_n with the understanding that they hold factors of c . Next we notice that in the linearized theory, once the background cosmological constants are set to zero, the β_n parameters are only present in the Fierz-Pauli mass definition. Let β be the specific combination appearing in the mass definition, i.e.

$$\beta = \beta_1 + 2\beta_2 + \beta_3. \quad (2.27)$$

The cosmological constants are given by

$$\Lambda_g(c) = \frac{m^4}{m_g^2} (\beta_0 + 3\beta_1 + 3\beta_2 + \beta_3), \quad (2.28a)$$

$$\Lambda_f(c) = \frac{m^4}{(m_f)^2} (\beta_1 + 3\beta_2 + 3\beta_3 + \beta_4). \quad (2.28b)$$

A possible choice of the parameters β_n for which these constants are null is:

$$\beta_0 = \beta_4 = 0, \quad \beta_1 = \beta_3 = -\frac{3}{2}\beta, \quad \beta_2 = 6\beta. \quad (2.29)$$

The Fierz-Pauli mass term is given by

$$\bar{m}_{FP}^2 = \frac{m^4}{m_g^2} \frac{1}{\alpha^2} \beta. \quad (2.30)$$

This kind of parameter choice is called 1 parameter theory, because each β_n can be determined from the coefficient β . Furthermore, given that equation 2.24 links the Fierz-Pauli mass to β , and if we suppose that the interaction scale m is a constant, then the theory could be studied in terms of the parameters α and m_{FP} . Henceforward, we suppose that m_{FP} and α are independent and the results are given in terms of these variables. Imposing the relations 2.29 on the parameters of the theory ensures the existence of flat background solutions. Thus we take $\bar{G}_{\mu\nu} = \eta_{\mu\nu}$, the mostly positive flat Minkowski metric.

With flat background solutions at hand, we can now define the physical situation for which we propose to calculate the time delay: We consider 2 massless spin-0 particles that couple only to the metric $g_{\mu\nu}$. The reason the coupling is restricted to one metric is that this is the only known case that guarantees that the ghost-free nature of the theory is preserved. Another possible choice would be that one boson couples to $g_{\mu\nu}$ while the other couples to $f_{\mu\nu}$. The action for this coupling is given by

$$S = -\frac{1}{2} \sum_{a=1}^2 \int d^4x \sqrt{-g} g^{\mu\nu} \partial_\mu \phi_a \partial_\nu \phi_a. \quad (2.31)$$

Recalling that $g^{\mu\nu} = \bar{g}^{\mu\nu} + \delta g^{\mu\nu}$, and the choice $\bar{G}_{\mu\nu} = \eta_{\mu\nu}$, we get

$$g_{\mu\nu} = \frac{1}{1 + \alpha^2} \eta_{\mu\nu} + \delta g_{\mu\nu}. \quad (2.32)$$

Using this form we can express $\sqrt{-g}$ to first order in the perturbation as

$$\sqrt{-g} = \sqrt{-\bar{g}} \left(1 - \frac{1}{2} \delta g^{\mu\nu} \bar{g}_{\mu\nu} \right). \quad (2.33)$$

Using $\bar{g}_{\mu\nu} = \frac{1}{1 + \alpha^2} \eta_{\mu\nu}$ we get

$$\sqrt{-g} = \frac{1}{(1 + \alpha^2)^2} \left(1 - \frac{1}{2} \delta g^{\mu\nu} \bar{g}_{\mu\nu} \right). \quad (2.34)$$

Consider next the integrand in the action 2.31, $I_a = \sqrt{g} g^{\mu\nu} \partial_\mu \phi_a \partial_\nu \phi_a$, the full expansion

leads to

$$I_a = \frac{1}{(1+\alpha^2)^2} \left(1 - \frac{1}{2} \frac{1}{(1+\alpha^2)} \delta g^{\mu\nu} \eta_{\mu\nu}\right) ((1+\alpha^2) \eta^{\rho\sigma} + \delta g^{\rho\sigma}) \partial_\rho \phi_a \partial_\sigma \phi_a, \quad (2.35a)$$

$$I_a = \frac{1}{(1+\alpha^2)} \partial^2 \phi_a + \frac{1}{(1+\alpha^2)^2} \delta g^{\rho\sigma} (\partial_\rho \phi_a \partial_\sigma \phi_a - \frac{1}{2} \eta_{\rho\sigma} \partial^2 \phi_a). \quad (2.35b)$$

In the second line, we only kept terms linear in the perturbation and $\partial^2 \phi_a = \eta^{\mu\nu} \partial_\mu \phi_a \partial_\nu \phi_a$. The action 2.31 could then be decomposed into a kinetic part,

$$S_K = -\frac{1}{2(1+\alpha^2)} \sum_{a=1}^2 \int d^4x \partial^2 \phi_a, \quad (2.36)$$

and an interaction part

$$S_I = -\frac{1}{2(1+\alpha^2)^2} \sum_{a=1}^2 \int d^4x \delta g^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a). \quad (2.37)$$

However, to ensure the consistency of the theory and the application of canonical quantization we redefine the fields to $\phi'_a = \frac{1}{(1+\alpha^2)^{\frac{1}{2}}} \phi_a$. Then we drop the ' notation. Notice also that when we introduced the perturbations $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$, we did not include the normalization factors in the definitions. Thus, in the canonical normalization we have

$$\delta g_{\mu\nu} = \frac{\delta g'_{\mu\nu}}{m_g}, \quad (2.38a)$$

$$\delta f_{\mu\nu} = \frac{\delta f'_{\mu\nu}}{m_f}. \quad (2.38b)$$

$$(2.38c)$$

Hence for the interaction action, a factor of $\frac{1}{m_g}$ should be introduced for consistency with the canonical quantization. Thus,

$$S_K = -\frac{1}{2} \sum_{a=1}^2 \int d^4x \partial^2 \phi_a, \quad (2.39)$$

and

$$S_I = -\frac{1}{2m_g(1+\alpha^2)} \sum_{a=1}^2 \int d^4x \delta g^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a). \quad (2.40)$$

Using equations 2.22, we can express the perturbation $\delta g_{\mu\nu}$ in terms of the massive

$\delta M_{\mu\nu}$ and massless modes $\delta G_{\mu\nu}$.

$$\delta g_{\mu\nu} = \frac{1}{1 + \alpha^2} (\delta G_{\mu\nu} - 2\alpha^2 \delta M_{\mu\nu}). \quad (2.41)$$

Using $\delta g^{\mu\nu} = -\bar{g}^{\mu\rho} \bar{g}^{\nu\sigma} \delta g_{\rho\sigma}$, we get

$$\delta g^{\mu\nu} = -(1 + \alpha^2)^2 \eta^{\mu\rho} \eta^{\nu\sigma} \delta g_{\rho\sigma}, \quad (2.42a)$$

$$= -(1 + \alpha^2) \eta^{\mu\rho} \eta^{\nu\sigma} (\delta G_{\rho\sigma} - 2\alpha^2 \delta M_{\rho\sigma}), \quad (2.42b)$$

$$= -(1 + \alpha^2) (\delta G^{\mu\nu} - 2\alpha^2 \delta M^{\mu\nu}). \quad (2.42c)$$

Thus we can rewrite the interaction action as

$$\begin{aligned} S_I = \sum_{a=1}^2 \int d^4x \frac{-1}{2m_g} \delta G^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a) + \\ \frac{\alpha^2}{m_g} \delta M^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a). \end{aligned} \quad (2.43)$$

The total action describing the interaction of two massless scalar bosons with the massive and massless modes is given by (the bar notation has been drop):

$$S = S_{KG} + S_{KM} + S_K + S_{IG} + S_{IM}, \quad (2.44)$$

where

$$S_{KG} = \frac{1}{2} \int d^4x \delta G^{\mu\nu} \varepsilon_{\mu\nu}{}^{\rho\sigma} \delta G_{\rho\sigma}, \quad (2.45a)$$

$$S_{KM} = \frac{1}{2} \int d^4x \delta M^{\mu\nu} [\varepsilon_{\mu\nu}{}^{\rho\sigma} \delta M_{\rho\sigma} + \frac{m_{FP}^2}{2} (\delta M_{\mu\nu} - \eta_{\mu\nu} \delta M)], \quad (2.45b)$$

$$S_K = -\frac{1}{2} \sum_{a=1}^2 \int d^4x \partial^2 \phi_a, \quad (2.45c)$$

$$S_{IG} = \sum_{a=1}^2 \int d^4x -\frac{1}{2m_g} \delta G^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a), \quad (2.45d)$$

$$S_{IM} = \sum_{a=1}^2 \int d^4x \frac{\alpha^2}{m_g} \delta M^{\mu\nu} (\partial_\mu \phi_a \partial_\nu \phi_a - \frac{1}{2} \eta_{\mu\nu} \partial^2 \phi_a). \quad (2.45e)$$

From the expression of the action one can observe that the matter fields ϕ_a couple to both the massive and massless modes but with different coupling constant. Let γ_M and γ_G be the coupling constant for the massive and massless modes respectively. The ratio between these constants depends only on $\alpha = \frac{m_f}{m_g}$ and is given by

$$\frac{\gamma_M}{\gamma_G} = -2\alpha^2 \quad (2.46)$$

In [23], the authors argue that for the bimetric theory to be in accordance with GR on large scale, the parameter α should be small. Taking this constraint, we deduce that the interaction should be dominated by the massless mode and the massive graviton constitute only a correction.

From the expression of action 2.45 the Feynman rules for the interaction are deduced:

1. The propagator for the massive mode is

$$D_{\mu\nu,\rho\sigma}^{(m)}(p) = \frac{-i}{p^2 + m_{FP}^2 - i\epsilon} \left\{ \frac{1}{2} (P_{\mu\rho} P_{\nu\sigma} + P_{\mu\sigma} P_{\nu\rho}) - \frac{1}{3} P_{\mu\nu} P_{\rho\sigma} \right\}, \quad (2.47)$$

With

$$P_{\mu\nu} = \eta_{\mu\nu} - \frac{p_\mu p_\nu}{m_{FP}^2}. \quad (2.48)$$

2. The propagator for the massless mode is:

$$D_{\mu\nu,\rho\sigma}^{(0)}(p) = \frac{-i}{2(p^2 - i\epsilon)} (\eta_{\mu\rho} \eta_{\nu\sigma} + \eta_{\mu\sigma} \eta_{\nu\rho} - \eta_{\mu\nu} \eta_{\rho\sigma}). \quad (2.49)$$

3. The propagators for the massless scalars are:

$$G_a(k) = \frac{-i}{k^2 - i\epsilon}. \quad (2.50)$$

where the subscript a indicates the type of the boson (1 or 2).

4. The vertex coefficient for the massless mode is given by

$$\tau_{(0)}^{\mu\nu}(p, p') = \frac{i}{2m_g} (p^\mu p'^\nu + p^\nu p'^\mu - \eta^{\mu\nu} (p \cdot p')). \quad (2.51)$$

5. Finally, the vertex coefficient for the massive mode is

$$\tau_{(m)}^{\mu\nu} = -2\alpha^2 \tau_{(0)}^{\mu\nu}. \quad (2.52)$$

2.2 Gravitational Scattering Amplitude in the Eikonal Regime

With the Feynman rules at hand, we can proceed to calculate scattering amplitudes. In this section we follow the steps in [22] to calculate the eikonal phase to first order

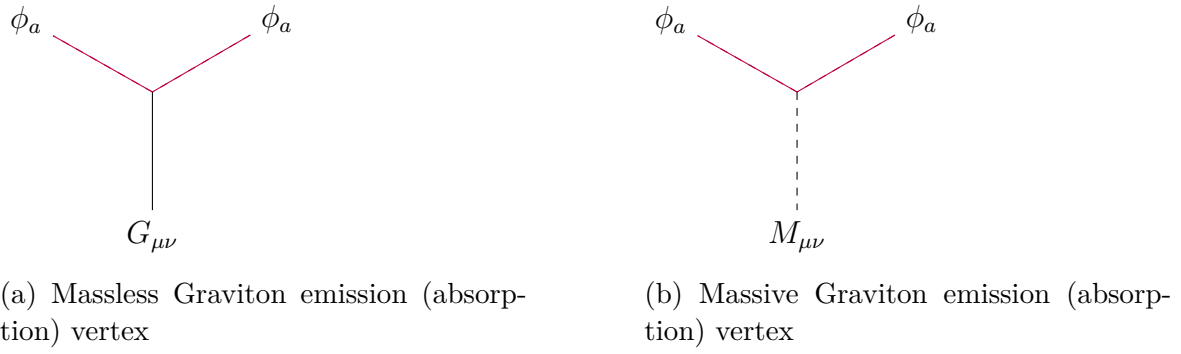


Figure 2.1: Possible vertices in the theory of scalar particles interaction through bimetric gravity.

and see how time delay is then derived.

Consider the case where the massless bosons ϕ_1 and ϕ_2 interact gravitationally. Given the structure of the vertices in the theory, starting from an initial state of 2 spin-0 particles of different types, the only possible interaction channel is the t -channel. We propose to study this interaction within the eikonal regime, where the center of mass energy $\sqrt{s}$ is much larger than the momentum transfer between the particles. Physically this corresponds to highly energetic particles scattering weakly through gravity. This corresponds to the following power scale

$$|p_1 \cdot p_2| \gg |q^2|. \quad (2.53)$$

Here, q refers to any interchanged momenta between the 2 particles. In the diagrams that follow we use the convention:

1. Blue lines for scalar bosons of type 1
2. Red lines for scalar bosons of type 2
3. Black continuous line for massless graviton
4. Black dashed lines for massive graviton

2.2.1 Massless graviton exchange

Consider first the tree level t channel interaction between the 2 bosons:

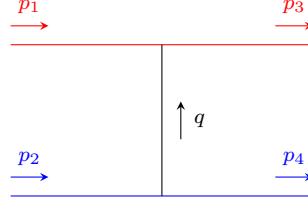


Figure 2.2: Tree level Feynman diagram for the scattering of ϕ_1 and ϕ_2 by the exchange of massless graviton

The vertex factor $\tau_{(0)}^{\mu\nu}(p_1, p_1 + q)$ is

$$\tau_{(0)}^{\mu\nu}(p_1, p_1 - q) = \frac{i}{2m_g} \{p_1^\mu (p_1 - q)^\nu + p_1^\nu (p_1 - q)^\mu - (p_1 \cdot (p_1 - q))\}, \quad (2.54a)$$

$$\tau_{(0)}^{\mu\nu}(p_1, p_1 + q) = \frac{i}{2m_g} \{2p_1^\mu p_1^\nu - p_1^\mu q^\nu - p_1^\nu q^\mu + p_1 \cdot q\}. \quad (2.54b)$$

and that we consider the momentum transfer to be negligible with respect to the center of mass energy, we can neglect the terms where q and p_1 mix. Thus, to first order the massless vertex interaction term is given by

$$\tau_{(0)}^{\mu\nu}(p_1, p_1 - q) \approx \frac{i}{m_g} p_1^\mu p_1^\nu. \quad (2.55)$$

Therefore,

$$\tau_{(0)}^{\mu\nu}(p_1, p_3) D_{\mu\nu, \rho\sigma}^{(0)}(q) \tau_{(0)}^{\mu\nu}(p_2, p_4) = \frac{i}{2m_g^2(q^2 - i\epsilon)} p_1^\mu p_1^\nu (\eta_{\mu\rho} \eta_{\nu\sigma} + \eta_{\mu\sigma} \eta_{\nu\rho} - \eta_{\mu\nu} \eta_{\rho\sigma}) p_2^\rho p_2^\sigma, \quad (2.56a)$$

$$\tau_{(0)}^{\mu\nu}(p_1, p_3) D_{\mu\nu, \rho\sigma}^{(0)}(q) \tau_{(0)}^{\mu\nu}(p_2, p_4) = \frac{i}{m_g^2(q^2 - i\epsilon)} (p_1 \cdot p_2)^2. \quad (2.56b)$$

Recalling the expression for s

$$s = -(p_1 + p_2)^2 = -2(p_1 \cdot p_2). \quad (2.57)$$

we finally get

$$\tau_{(0)}^{\mu\nu}(p_1, p_3) D_{\mu\nu, \rho\sigma}^{(0)}(q) \tau_{(0)}^{\mu\nu}(p_2, p_4) = \frac{is^2}{4m_g^2 q^2}. \quad (2.58)$$

Let $A_1^0(s, q)$ be the amplitude of the scattering of the ϕ_1 and ϕ_2 bosons due to the

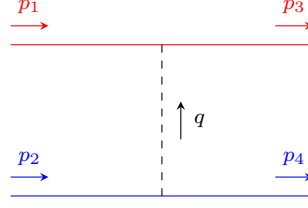


Figure 2.3: t channel Feynman diagram for the scattering of ϕ_1 and ϕ_2 by the exchange of massive graviton.

exchange of one massless graviton. Then we get

$$iA_1^0(s, q) = \frac{is^2}{4m_g^2 q^2}, \quad (2.59a)$$

$$= \frac{is^2 \gamma_G^2}{q^2}. \quad (2.59b)$$

2.2.2 Massive graviton exchange

We next consider the same process as in the previous section but this time the interaction is due to the exchange of the massive graviton.

Using the relation between the vertices for the exchange of massive and massless modes 2.52, we get

$$\tau_m^{\mu\nu}(p_1, p_1 - q) = \frac{-2i\alpha^2}{m_g} p_1^\mu p_1^\nu. \quad (2.60)$$

Therefore

$$\tau_m^{\mu\nu}(p_1, p_3) D_{\mu\nu, \rho\sigma}^{(m)}(q) \tau_m^{\mu\nu}(p_2, p_4) = \frac{4i\alpha^4}{m_g^2(q^2 + m_{FP}^2)} p_1^\mu p_1^\nu p_2^\rho p_2^\sigma \left\{ \frac{1}{2}(Q_{\mu\rho} Q_{\nu\sigma} + Q_{\mu\sigma} Q_{\nu\rho}) - \frac{1}{3} Q_{\mu\nu} Q_{\rho\sigma} \right\}. \quad (2.61)$$

The term $p_1^\mu p_1^\nu p_2^\rho p_2^\sigma Q_{\mu\rho} Q_{\nu\sigma}$ leads to

$$p_1^\mu p_1^\nu p_2^\rho p_2^\sigma Q_{\mu\rho} Q_{\nu\sigma} = p_1^\mu p_1^\nu p_2^\rho p_2^\sigma \left(\eta_{\mu\rho} - \frac{q_\mu q_\rho}{m_{FP}^2} \right) \left(\eta_{\nu\sigma} - \frac{q_\nu q_\sigma}{m_{FP}^2} \right), \quad (2.62a)$$

$$= (p_1 \cdot p_2)^2 - 2 \frac{(p_1 \cdot q)(p_2 \cdot q)(p_1 \cdot p_2)}{m_{FP}^2} + \frac{(p_1 \cdot q)^2 (p_2 \cdot q)^2}{m_{FP}^4}, \quad (2.62b)$$

$$= \frac{s^2}{4} - \frac{st^2}{4m_{FP}^2} + \frac{t^4}{16m_{FP}^4}, \quad (2.62c)$$

$$\approx \frac{s^2}{4}. \quad (2.62d)$$

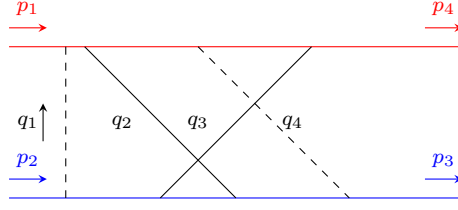


Figure 2.4: Feynman diagram for the scattering of ϕ_1 and ϕ_2 by the exchange of $n = 4$ gravitons.

Similarly, we can show that

$$p_1^\mu p_1^\nu p_2^\rho p_2^\sigma Q_{\mu\nu} Q_{\rho\sigma} = \frac{-t^4}{16m_{FP}^4}. \quad (2.63)$$

Thus to leading order in s we get,

$$\tau_m^{\mu\nu}(-p_1, p_4) D_{\mu\nu, \rho\sigma}^{(m)}(q) \tau_m^{\mu\nu}(-p_2, -p_2 + q) = \frac{i\alpha^4 s^2}{m_g^2(q^2 + m_{FP}^2)}. \quad (2.64)$$

Let $A_1^m(s, q)$ be the amplitude for the scattering of ϕ_1 and ϕ_2 in the t channel due to the exchange of one massive graviton. We then have

$$iA_1^m(s, q) = \frac{i\alpha^4 s^2}{m_g^2(q^2 + m_{FP}^2)}, \quad (2.65a)$$

$$= \frac{i\gamma_M^2 s^2}{(q^2 + m_{FP}^2)}. \quad (2.65b)$$

2.2.3 General Ladder diagram and the eikonal phase:

In the previous 2 subsections, we derived the amplitude for the tree level scattering of the 2 boson flavors in the t channel, considering first the exchange of a massless graviton and in the second the exchange of massive one. In this section we propose to study a general diagram where the two bosons exchange a total of n gravitons that can be massive or massless. The sum of the amplitudes for all the possible Feynman diagrams with n gravitons exchanges is noted by $A_n(s, q)$ where q denotes the total momentum exchanged between the 2 bosons. A possible Feynman diagram for $n = 4$ is given by figure 2.4. For a given n , the amplitude $A_n(s, q)$ takes the form:

$$iA_n(s, q) = \sum_{k=0}^n C_n^k (\gamma_G^2 s^2)^k (\gamma_M^2 s^2)^{n-k} \int \left[\prod_{i=1}^k \left(\frac{-i}{q_i^2 - i\epsilon} \right) \frac{d^4 q_i}{(2\pi)^4} \right] \int \left[\prod_{j=1}^{n-k} \left(\frac{-i}{q_j^2 + m_{FP}^2} \right) \frac{d^4 q_j}{(2\pi)^4} \right] \delta^{(4)}(q - \sum_{i=1}^n q_i) I^{(n)}. \quad (2.66)$$

C_n^k is the binomial coefficient, i.e. $C_n^k = \frac{n!}{k!(n-k)!}$.

This formula can be understood as following:

1. For a given n , we sum over all the possible combination of k massless gravitons exchange and $n - k$ massive ones. This explains the sum over k and the binomial factor that encodes the different ways a set of k elements could be chosen from a set of n elements.
2. $(\gamma_G^2 s^2)^k$ and $(\gamma_M^2 s^2)^{n-k}$ correspond to the vertex factors due to the exchanges of k massless gravitons and $n - k$ massive ones.
3. The product terms, corresponds to the propagators of the gravitons where we integrate all the possible momenta.
4. The delta functions ensures conservation of energy momentum
5. $I^{(n)}$ encodes all ladder topologies of suitable product of the massless bosons propagators and their contractions with the suitable massless and massive gravitons vertices and propagators. In other words $I^{(n)}$ sums over the different orders the momenta q_i of massless and massive gravitons are emitted and absorbed and internal propagators of the massless bosons that appear.

Explicitly, $I^{(n)}$ are given by

$$I^{(0)} = 1, \tag{2.67a}$$

$$I^{(n)} = \frac{1}{n!} \sum_{\sigma \in S_n} G_1(p_1 + q_{\sigma_1}) \dots G_1(p_1 + q_{\sigma_1} + \dots + q_{\sigma_{n-1}}) \cdot \sum_{\sigma' \in S_n} G_2(p_2 + q_{\sigma'_1}) \dots G_2(p_2 + q_{\sigma'_1} + \dots + q_{\sigma'_{n-1}}). \tag{2.67b}$$

G_1 and G_2 are the propagators of the massless bosons 2.50, and S_n is the permutation group of n elements. The factor $\frac{1}{n!}$ is to average over topologically identical diagrams.

To advance further, we consider the following change of variables:

$$q^\mu = \frac{1}{s}(q_1 p_1^\mu + q_2 p_2^\mu) + q_\perp^\mu, \tag{2.68}$$

with $q_1 = 2p_2 \cdot q$ and $q_2 = 2p_1 \cdot q$. $q_\perp$ orthogonal to p_1 and p_2 .

$$dq^2 = -\frac{1}{s} dq_1 dq_2 + dq_\perp^2. \tag{2.69}$$

The Jacobian corresponding to this variable transformation is given by $\frac{1}{2s}$. Hence, we get

$$\delta^{(4)}(q - \sum_{i=1}^n q_i) = 2s \delta(q_1 - \sum_{i=1}^n q_{i1}) \delta(q_2 - \sum_{i=1}^n q_{i2}) \delta^{(2)}(q_{\perp} - \sum_{i=1}^n q_{i\perp}), \quad (2.70a)$$

$$\approx 2s \delta(\sum_{i=1}^n 2p_1 \cdot q_i) \delta(\sum_{i=1}^n 2p_2 \cdot q_i) \delta^{(2)}(q_{\perp} - \sum_{i=1}^n q_{i\perp}). \quad (2.70b)$$

$$(2.70c)$$

In the last equation we have neglected terms q_1 and q_2 because they are of the order $q^2 = t$, and thus can be neglected in our development to first order in s . We now focus on the propagators of the massless bosons. To first order we get

$$G_1(p_1 + q_{\sigma}) = \frac{-i}{(p_1 + q_{\sigma})^2 - i\epsilon}, \quad (2.71a)$$

$$= \frac{-i}{2p_1 \cdot q_{\sigma} + q_{\sigma}^2 - i\epsilon}, \quad (2.71b)$$

$$\approx \frac{-i}{2p_1 \cdot q_{\sigma} - i\epsilon}. \quad (2.71c)$$

$$(2.71d)$$

Let $\omega_{\sigma}^1 = 2p_1 \cdot q_{\sigma}$ and $\omega_{\sigma}^2 = -2p_2 \cdot q_{\sigma}$. We also define

$$f(a_1, \dots, a_n) = \sum_{\sigma \in S_n} \prod_{k=1}^n \left(\sum_{j=1}^k a_{\sigma_j} \right)^{-1}. \quad (2.72)$$

With this definition we can write:

$$\begin{aligned} \delta^{(4)}(q - \sum_{i=1}^n q_i) I^{(n)} &= 2s \delta^{(2)}(q_{\perp} - \sum_{i=1}^n q_{i\perp}) \delta(\sum_{i=1}^n \omega_i^1) \delta(\sum_{i=1}^n \omega_i^2) \\ &\quad \frac{(-i)^{2(n-1)}}{n!} f(w_1^1, \dots, w_n^1) f(w_1^2, \dots, w_n^2). \end{aligned} \quad (2.73)$$

Using the result from [22] we have

$$\delta(\sum_{i=1}^n \omega_i^1) f(w_1^1, \dots, w_n^1) = (2i\pi)^{n-1} \prod_{i=1}^n \delta(w_i^1). \quad (2.74)$$

We can finally rewrite the general amplitude 2.66 as

$$\begin{aligned}
iA_n(s, q) &= 2s \frac{1}{n!} \sum_{k=0}^n C_n^k (\gamma_G^2 s^2)^k (\gamma_M^2 s^2)^{n-k} (2\pi)^2 \delta^{(2)}(q_\perp - \sum_{i=1}^n q_{\perp i}) \\
&\quad \int \left[\prod_{i=1}^k \left(\frac{-i}{q_i^2 - i\epsilon} \right) \delta(2p_1 \cdot q_i) \delta(2p_2 \cdot q_i) \frac{d^4 q_i}{(2\pi)^2} \right] \\
&\quad \int \left[\prod_{j=1}^{n-k} \left(\frac{-i}{q_j^2 + m_{FP}^2} \right) \delta(2p_1 \cdot q_j) \delta(2p_2 \cdot q_j) \frac{d^4 q_j}{(2\pi)^2} \right].
\end{aligned} \tag{2.75}$$

The final step is to define the Fourier transform to the impact parameter space

$$FT[A](s, b) = \int e^{ib \cdot q} (2\pi) \delta(2p_1 \cdot q) (2\pi) \delta(2p_2 \cdot q) A(s, -q^2) \frac{d^4 q}{(2\pi)^4}, \tag{2.76a}$$

$$= \int e^{ib \cdot q_\perp} \frac{A(s, -q_\perp^2)}{2s} \frac{d^2 q_\perp}{(2\pi)^2}. \tag{2.76b}$$

Transforming the general amplitude, to the impact parameter space we get

$$iFT[A_n](s, b) = \frac{1}{n!} (iFT[A_0^0](s, b) + iFT[A_0^m](s, b))^n, \tag{2.77}$$

where $FT[A_0^0](s, b)$ and $FT[A_0^m](s, b)$ are the Fourier transforms to impact parameter space of the tree level process with massless and massive graviton exchange respectively. Finally, using the same normalization as [22], the scattering matrix element for this process is given by

$$\tilde{S}(s, b) = 1 + \sum_{n=1}^{+\infty} iFT[A_n](s, b), \tag{2.78a}$$

$$= 1 + \frac{1}{n!} \sum_{n=1}^{+\infty} (iFT[A_0^0](s, b) + iFT[A_0^m](s, b))^n, \tag{2.78b}$$

$$= e^{2i\delta(s, b)}, \tag{2.78c}$$

where

$$2i\delta(b, s) = iFT[A_0^0](s, b) + iFT[A_0^m](s, b). \tag{2.79}$$

Explicitly, we get

$$2i\delta(b, s) = \frac{is}{m_g^2} \int e^{ib \cdot q_\perp} \left(\frac{1}{4q_\perp^2} + \frac{\alpha^4}{q_\perp^2 + m_{FP}^2} \right) \frac{d^2 q_\perp}{(2\pi)^2}. \tag{2.80}$$

$\delta(s, b)$ is the eikonal phase.

Before moving forward with our analysis, we need to take few steps back and justify why

the ladder diagrams 2.4 are leading in the eikonal approximation. Here we reproduce the argument proposed in [22]: During the construction of the linearized interaction theory, we developed $\sqrt{-g}$ to only first order in the perturbation $\delta g_{\mu\nu}$, yet higher order terms would have lead to diagrams where one spin-0 particle emits or absorbers multiple gravitons within the same vertex. Thus for a given number of a gravitons exchanged between the 2 spinless particles, we could have different diagrams where a certain number of gravitons are emitted or absorbed from the same vertex. Yet such diagrams have not been considered in the eikonal summation. The reason they can be neglected in the eikonal approximation is best illustrated with a diagram.

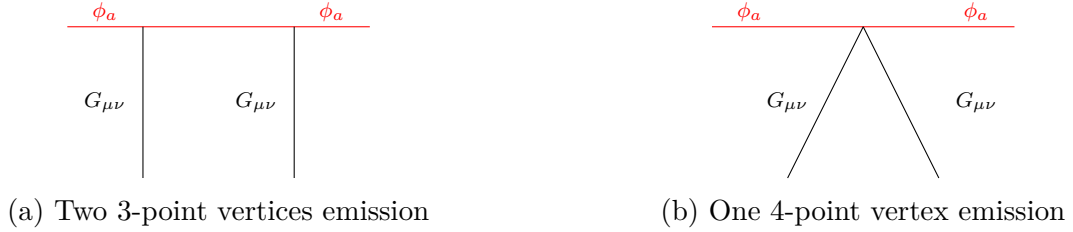


Figure 2.5: Comparison between the emission of two gravitons from two different three-point vertices (2.5a) and from one four-point vertex (2.5b). In the first case each vertex contributes a factor $\gamma_g s$ to the amplitude; the energetic scalar propagator scales as $1/\sqrt{s}$, so the total contribution is $\mathcal{O}(\gamma_g^2 s^{3/2})$. In the second case the four-point vertex contributes $\gamma_g^2 s$. Each time a factor of $\sqrt{s}$ is lost, a factor of q is needed, so four-point vertices are subleading.

While the 2 diagrams 2.5a and 2.5b are of the same order in the coupling constant, they have different scaling in the center of mass energy. For ladder diagrams each vertex contribute a factor of $\gamma_g s$ to the total amplitude while the energetic scalar propagator contribute a factor $1/\sqrt{s}$ resulting in a total contribution of $\mathcal{O}(\gamma_g^2 s^{3/2})$. In the second case, the vertex scales as $\gamma_g^2 s$. Thus we lose a factor of $\sqrt{s}$ that is compensated with q . Therefore ladder diagrams are leading.

Compared to the result in [22], we see that the contribution from the massive mode interaction is suppressed by the factor $\frac{\gamma_M}{\gamma_g} = -2\alpha^2$ squared. Most of the measurable features of the interaction under study are directly derived from the eikonal phase, and thus one concludes that, if the factor α is taken to be small, such that the bimetric theory stays fairly close to the well tested GR, the correction from the massive mode channel are suppressed by $4\alpha^4$ factor.

2.2.4 Time delay calculation

With the expression of the eikonal phase at hand, the next step is to calculate the time delay experienced by the two scalar boson due to the gravitational interaction. Following [22], this is done by transforming the scattering matrix element $\tilde{S}(s, b)$ in the time domain by applying a Fourier transform over the center of mass energy $\sqrt{s}$. Thus, we get

$$\tilde{\hat{S}}(t, b) = \int d(\sqrt{s}) e^{-i\sqrt{s}t} \tilde{S}(s, b), \quad (2.81a)$$

$$= \int d(\sqrt{s}) e^{-i\sqrt{s}t} e^{2i\delta(s, b)}, \quad (2.81b)$$

$$= \int d(\sqrt{s}) e^{i(2\delta(s, b) - \sqrt{s}t)}. \quad (2.81c)$$

$$(2.81d)$$

The time delay is then determined through the stationary phase method and we get

$$\Delta T(s, b) = \frac{\partial(2\delta(s, b))}{\partial\sqrt{s}}, \quad (2.82a)$$

$$= \frac{\sqrt{s}}{2m_g^2\pi^2} \int e^{ib \cdot q_\perp} \left(\frac{1}{4q_\perp^2} + \frac{\alpha^4}{q_\perp^2 + m_{FP}^2} \right) d^2q_\perp. \quad (2.82b)$$

To express the time delay in a simpler form, we start by noting that in the center of mass energy and due to the metric convention we have

$$p_1 = \begin{pmatrix} \frac{E}{2} \\ \vec{p} \end{pmatrix}, \quad (2.83a)$$

$$p_2 = \begin{pmatrix} \frac{E}{2} \\ -\vec{p} \end{pmatrix}. \quad (2.83b)$$

So that the conditions $q_\perp \cdot p_1 = q_\perp \cdot p_2 = 0$ lead to

$$-q_{\perp 0} \frac{E}{2} + \vec{q}_\perp \cdot \vec{p} = 0, \quad (2.84a)$$

$$-q_{\perp 0} \frac{E}{2} - \vec{q}_\perp \cdot \vec{p} = 0. \quad (2.84b)$$

Which leads to $q_{\perp 0} = 0$ and $\vec{q}_{\perp} \cdot \vec{p} = 0$. Hence $q_{\perp}$ is space-like and transverse to the 3-momentum of the boson ϕ_1 and ϕ_2 . Then if we suppose that

$$\vec{p} = \begin{pmatrix} 0 \\ 0 \\ p \end{pmatrix}, \quad (2.85)$$

then we get

$$\vec{q}_{\perp} = \begin{pmatrix} q_{\perp x} \\ q_{\perp y} \\ 0 \end{pmatrix}. \quad (2.86)$$

Hence, the time delay can then be rewritten in cylindrical coordinates

$$\Delta T(s, b) = \frac{\sqrt{s}}{2m_g^2\pi^2} \int_0^{2\pi} \int_0^{\Lambda} e^{ibq_{\perp} \cos \theta} \left(\frac{1}{4q_{\perp}} + \frac{\alpha^4 q_{\perp}}{q_{\perp}^2 + m_{FP}^2} \right) dq_{\perp} d\theta. \quad (2.87)$$

Where Λ is the maximum momentum transfer. The introduction of this parameter is necessary because the eikonal approximation only holds if the momentum transfer is smaller than the center of mass energy.

The total time delay can be decomposed to a contribution from the massless mode that reproduces the known classical result of Shapiro time delay in GR (see appendix B for a proof using Bessel functions proprieties) and a contribution from the massive graviton mode.

2.3 Chapter Conclusion

In this chapter, we established the theoretical framework required to study gravitational time delay in bimetric gravity. We introduced the HRBG action, derived the equations of motion, and identified proportional background solutions that allow for a consistent linearization of the theory. This led to a natural decomposition of the metric perturbations into a massless and a massive spin-2 mode, each with well-defined propagation and interaction properties.

We then derived the matter couplings and corresponding Feynman rules for scalar fields interacting gravitationally, and computed the high-energy scattering amplitude in the eikonal regime. By summing the dominant ladder diagrams we obtained the eikonal phase, which directly determines the leading-order time delay experienced by massless scalar fields.

The results of this chapter provide the foundation for the phenomenological applications explored in the next chapter, where we use these tools to analyze photon propagation in the solar system and compare the resulting frequency shift to observational data.

Chapter 3

Constraints on Bimetric Gravity from Solar System Time Delay Measurements

In this chapter, we use time delay calculations in HRBG and the measurements from Cassini space mission to derive a relation between the Fierz Pauli mass m_{FP} and the ratio of Planck masses α so that time delay is within the experimentally allowed region. We start by conducting a literature overview of the constraints on HRBG from cosmological and solar system measurements. Then we present the modification needed to adapt the time delay calculations from the previous chapter to model the solar system. Finally, we compute the gravitational frequency shift, the observable measured in the mission, then use the constraints to derive the relationship.

3.1 Literature Review

1. Cosmological Constraints

Cosmological observations provide crucial constraints on bimetric gravity through their sensitivity to the expansion history and structure formation. [24] utilized Type Ia supernovae (SN1a) data from the Pantheon sample to constrain the physical parameter space of bimetric theory. The work spans multiple cosmological models depending on the number of non-zero β parameters of the theory. The methodology involved parameterizing the HRBG with an effective cosmological constant, the Fierz-Pauli mass m_{FP} of the massive graviton, and the ratio of Planck masses α . Then solving the modified Friedmann equations. Statistical

analysis using Markov Chain Monte Carlo (MCMC) method is then conducted to obtain posterior distributions for the bimetric parameters. The authors found that cosmological constraints are in agreement with different models of bimetric gravity. Especially that self-accelerating solution without cosmological constant are not excluded.

[25] employed a different approach by analyzing linear cosmological perturbations in bimetric gravity. They computed the growth factor of matter perturbations and compared it with Planck CMB data and large-scale structure surveys. The aim of the work was to set a limit on the allowed deviation from the Λ CMD model.

2. Solar System Tests

Solar System tests provide exceptionally stringent constraints due to the high precision of local gravitational measurements. [20] established the current gold standard for PPN parameter γ constraints through the Cassini spacecraft's superior conjunction experiment. Their methodology involved precise Doppler tracking of radio signals during solar conjunction, measuring the frequency shift caused by gravitational time delay. By fitting the time-dependent frequency shift pattern to the theoretical prediction, they obtained $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$. This measurement technique effectively probes the second-order spatial components of the metric, making it particularly sensitive to bimetric gravity modifications.

[26] investigated the Vainshtein screening mechanism in the solar system context. Their methodology focused on spherically symmetric solutions and the efficiency of screening at different radii. They computed the effective gravitational potential modifications and compared these with lunar laser ranging (LLR) data and planetary ephemerides. The key finding was that viable bimetric theories must exhibit efficient Vainshtein screening within the solar system, requiring specific relations between the interaction parameters β_n .

3. Gravitational Wave Constraints

The advent of gravitational wave astronomy has opened new avenues for testing bimetric gravity. [27] analyzed the modified dispersion relations for gravitational waves in bimetric theories. Their methodology involved computing the group velocity differences between massless and massive modes and comparing with GW170817 and GRB170817A observations. The nearly simultaneous arrival of gravitational and electromagnetic waves constrained the graviton mass to $m_g < 10^{-22}$ eV, though this primarily applies to the massless mode's propagation.

4. Methodological Comparison

The different methodological approaches reveal complementary strengths: cosmological tests probe the infrared modifications and expansion history, solar system tests provide high-precision constraints on local gravity, and gravitational wave observations test propagation effects. The PPN formalism employed in solar system tests offers the most stringent constraints on bimetric parameters, particularly α , due to the exceptional precision of Doppler tracking experiments like Cassini.

In this work, we build upon the Solar System methodology by implementing a direct calculation of the frequency shift observable within the eikonal approximation framework, providing a novel connection between quantum field theoretic techniques and high-precision experimental data. In this chapter we use the results of time delay calculation in bimetric theory and the restriction on deviation from general relativity result obtained through the Cassini spacecraft to restrict the parameters of the theory α and m_{FP} .

3.2 Cassini space mission and time delay calculation in the solar system

Cassini space mission was a joint interplanetary exploration project between NASA, the European space agency (ESA), and the Italian space agency. Its mission was to study the planet Saturn and its system. It was launched in 1997 and arrived on Saturn on 2004. It's mission lasted 13 years. The Cassini space mission provided another test for GR. In 2002, during a superior solar conjunction, the spacecraft was on the other side of the Sun with respect to earth. Radio frequency signals were sent from earth to the spacecraft and back. The time delay experienced by the signal was measured for different impact parameters and this was compared to GR prediction. The Cassini space mission provided strong constraints on any effect beyond general relativity [20]. This is expressed with the value of the Post Newtonian Parameter γ ,

$$\gamma - 1 = \frac{\Delta t_m}{\Delta t_{GR}} = (2.1 \pm 2.3) \cdot 10^{-5}. \quad (3.1)$$

γ measures the degree to which a theory diverges from general relativity predictions. For GR $\gamma = 1$.

The aim is to use this constraint to compute the valid parameter space of the bimetric

theory. However, our previous calculations cannot be applied to this situation. We calculated the time delay experienced by 2 massless spin-0 particles. In the case of the Cassini mission the time delay is due to the interaction of the photons with the Sun through gravity. While as a first approximation the spin of the photons could be ignored, the mass of the Sun can not be neglected. Thus, we propose to model the Sun as a spin-0 massive particle and the photons as spin-0 massless particles. This will have two impacts on our previous calculations:

1. The tree level amplitude will change.
2. In the derivation of the eikonal phase, we decomposed the gravitons momenta in orthogonal momenta $q_\perp$ with 2 components, the other 2 components were along the interacting spin-0 boson momenta. However this decomposition no longer work because $p_1^2 = -m_1^2$, so if we keep the same change of variable $q_\perp$ is no longer orthogonal to the incoming scalars momenta. In this case we loose the physical interpretation of b as impact parameter. This also results in the calculation of the Jacobian being altered. Thus, we need to find a new base in which we can perform a similar decomposing, where $q_\perp$ is orthogonal to the scalars incoming momenta.

3.2.1 Eikonal approximation in bimetric theory for a massive and massless spin-0 particles

In this section we summarize the important changes to the formalism developed in the previous chapter so that we can simulate the Cassini experiment.

Changes to the tree level amplitudes

Let θ and ϕ be the massless and massive spin-0 particles respectively representing a photon and the Sun. The addition of the mass of ϕ results in the following changes to the action:

$$S_{k_\phi} = -\frac{1}{2} \int d^4x (\partial^2 \phi + m_\phi^2 \phi^2), \quad (3.2a)$$

$$S_{IG_\phi} = \frac{1}{2m_g} \int d^4x \delta G^{\mu\nu} (\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} (\partial^2 \phi + m_\phi^2 \phi^2)), \quad (3.2b)$$

$$S_{IG_\phi} = -\frac{\alpha^2}{m_g} \int d^4x \delta M^{\mu\nu} (\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} \eta_{\mu\nu} (\partial^2 \phi + m_\phi^2 \phi^2)). \quad (3.2c)$$

Hence, we get the following changes in the Feynman rules:

$$G_\phi(p) = \frac{-i}{p^2 + m_\phi^2 - i\epsilon}, \quad (3.3a)$$

$$\tau_{(0)\phi}^{\rho\sigma}(p, p') = \frac{-i}{2m_g^2}(p^\rho p'^\sigma + p^\sigma p'^\rho - \eta^{\rho\sigma}(p \cdot p' + m_\phi^2)), \quad (3.3b)$$

$$\tau_{(m)\phi}^{\rho\sigma}(p, p') = \frac{i\alpha^2}{m_g^2}(p^\rho p'^\sigma + p^\sigma p'^\rho - \eta^{\rho\sigma}(p \cdot p' + m_\phi^2)). \quad (3.3c)$$

Where G_ϕ is the propagator for the massive spin-0 particle. $\tau_{(0)\phi}^{\rho\sigma}$ $\tau_{(m)\phi}^{\rho\sigma}$ are respectively the vertex coefficient, for an interaction of ϕ with the massless and massive gravitons. With these new Feynman rules we can again compute the tree level amplitudes. We find

$$A_1^0(s, q) = \frac{i(s - m_\phi)^2}{4m_g^2 q^2}, \quad (3.4a)$$

$$A_1^m(s, q) = \frac{i\alpha^4(s - m_\phi)^2}{m_g^2(q^2 + m_{FP}^2)}. \quad (3.4b)$$

Proof of consistency of the Fourier transformation to impact parameter space for massive and massless spin zero particle interaction

In the application of the eikonal approximation in the massless scalars case, the most important step in the summation was the use of equation 2.74. This required changing the coordinate system to those of 2.68. However, if we are to use the same coordinate system again $q_\perp$ would not be orthogonal to p_1 (4 momentum of ϕ) and p_2 (4 momentum of θ), and hence we can not interpret the Fourier transform 2.76 as a transformation to impact parameter space. Thus, for the consistency of the application of the eikonal approximation for the case of one massless and one massive spin-0 particles, it is essential to define a new coordinates system.

Let x_1^μ and x_2^μ be 2 vectors, we require the following conditions:

1. x_1^μ and x_2^μ are linear combinations of p_1^μ and p_2^μ .
2. $\forall$ 4-vector v^μ we can write $v^\mu = v_1 x_1^\mu + v_2 x_2^\mu + v_\perp^\mu$.
3. $v_1 = 2(p_2 \cdot v)$ and $v_2 = 2(p_1 \cdot v)$.
4. $v_\perp \cdot p_1 = v_\perp \cdot p_2 = 0$.

Let's suppose that x_1 and x_2 exist, then for $v^\mu = p_1^\mu$ and $v^\mu = p_2^\mu$ we get

$$p_1^\mu = 2(p_1 \cdot p_2)x_1^\mu + 2p_1^2 x_2 + p_{1\perp}, \quad (3.5a)$$

$$p_2^\mu = 2(p_1 \cdot p_2)x_2^\mu + p_{2\perp}. \quad (3.5b)$$

Taking the scalar product of the first and second equations with $p_{1\perp}$ and $p_{2\perp}$ respectively, we deduce from the first and second conditions that $p_{1\perp}$ and $p_{2\perp}$ are null vectors. The Mandelstam variable s is given by

$$s = -(p_1 + p_2)^2 = m_\phi^2 - 2p_1 \cdot p_2. \quad (3.6)$$

Hence, we can rewrite the system of equations 3.5 as

$$p_1^\mu = (m_\phi^2 - s)x_1^\mu - 2m_\phi^2 x_2^\mu, \quad (3.7a)$$

$$p_2^\mu = (m_\phi^2 - s)x_2^\mu. \quad (3.7b)$$

Inverting this relations we get

$$x_1^\mu = \frac{1}{(m_\phi^2 - s)}p_1^\mu + \frac{2m_\phi^2}{(m_\phi^2 - s)^2}p_2^\mu, \quad (3.8a)$$

$$x_2^\mu = \frac{1}{(m_\phi^2 - s)}p_2^\mu. \quad (3.8b)$$

It is now easy to verify that x_1^μ and x_2^μ verify the conditions mentioned previously.

These vectors verify the following properties:

$$x_1^2 = \frac{m_\phi^2}{(m_\phi^2 - s)^2}, \quad (3.9a)$$

$$x_2^2 = 0, \quad (3.9b)$$

$$x_1 \cdot x_2 = \frac{1}{2(m_\phi^2 - s)}. \quad (3.9c)$$

Thus, we get

$$dq^2 = \frac{m_\phi^2}{(m_\phi^2 - s)^2}dq_1^2 + \frac{1}{(m_\phi^2 - s)}dq_1 dq_2 + d^2 q_\perp. \quad (3.10)$$

The Jacobian of this transformation is $\frac{1}{4(m_\phi^2 - s)^2}$. Hence the volume element in this new coordinate system is

$$d^4 q = \frac{1}{2(s - m_\phi^2)}dq_1 dq_2 d^2 q_\perp. \quad (3.11)$$

We therefore conclude that the exponentiations procedures in the eikonal approxima-

tion is indeed applicable and we get

$$FT[A](s, q) = \int e^{ibq} (2\pi) \delta(2p_1 \cdot q) (2\pi) \delta(2p_2 \cdot q) A(s, -q^2) \frac{d^4 q}{(2\pi)^4}, \quad (3.12a)$$

$$= \frac{1}{2(s - m_\phi^2)} \int e^{ibq_\perp} A(s, -q_\perp^2) \frac{d^2 q_\perp}{(2\pi)^2}. \quad (3.12b)$$

The leading eikonal phase is therefore

$$2\delta_0(s, b) = FT[A_1^{(0)}](s, b) + FT[A_1^{(m)}](s, b), \quad (3.13a)$$

$$= \frac{(s - m_\phi^2)}{8m_g^2\pi^2} \int dq d\theta e^{ibq \cos(\theta)} \left(\frac{1}{4q} + \frac{\alpha^4 q}{q^2 + m_{FP}^2} \right). \quad (3.13b)$$

In the second line we expressed the integral in cylindrical coordinates. The time delay is therefore given by

$$\Delta T = \frac{\sqrt{s}}{4m_g^2\pi^2} \int_0^\Lambda \int_0^{2\pi} e^{ibq \cos(\theta)} \left(\frac{1}{4q} + \frac{\alpha^4 q}{q^2 + m_{FP}^2} \right) dq d\theta. \quad (3.14)$$

3.2.2 Model of the Cassini experiment and restriction on bimetric theory parameters

As mentioned before, we propose to use the Cassini experiment to find restriction on the parameter space in bimetric theory. We model the Sun with the spin zero particle ϕ with a mass $m_\phi = M_s$ and the photon with the particle θ . The spin of the photon is ignored.

The energy of the photon is negligible compared to that of the mass of the Sun and thus the center of mass energy of the system is equal to the mass of the Sun.

Hence, the time delay is given by

$$\Delta T = \frac{M_s}{4m_g^2\pi^2} \int_0^\Lambda dq \int_0^{2\pi} d\theta e^{ibq \cos(\theta)} \left(\frac{1}{4q} + \frac{\alpha^4 q}{q^2 + m_{FP}^2} \right). \quad (3.15)$$

In B, we prove that the massless-mode reproduces the correct GR time delay. As seen in B, the expression of the time delay as it is suffers from an infrared divergence due to the long range nature of massless gravity. This could be surmounted by computing the shift in the photon frequency due to the gravitational time delay. This is in fact the observable measured in [20]. The frequency shift is defined as the derivative of the

time delay with respect to time,

$$y_g = \frac{\partial \Delta T}{\partial t}, \quad (3.16a)$$

$$= \frac{\partial \Delta T}{\partial b} \frac{\partial b}{\partial t}. \quad (3.16b)$$

In [20], the authors approximate the variation of the impact parameter in time with the orbital velocity of earth. This is due to the large distance between the spacecraft and earth (8.5 AU) compared to the distance between earth and Sun (1 AU). The spacecraft could therefore be seen as stationary and the only relevant movement is that of the earth around the Sun. The frequency shift can then be decomposed to

$$y_G = \frac{\partial b}{\partial t} \frac{\partial \Delta t_G}{\partial b} = \frac{i M_s v_e}{16 m_g^2 \pi^2} \int_0^\Lambda \int_0^{2\pi} e^{ibq \cos(\theta)} \cos(\theta) dq d\theta, \quad (3.17a)$$

$$y_M = \frac{\partial b}{\partial t} \frac{\partial \Delta t_M}{\partial b} = \frac{i v_e M_s \alpha^4}{4 m_g^2 \pi^2} \int_0^\Lambda \int_0^{2\pi} e^{ibq \cos(\theta)} \frac{\cos(\theta) q^2}{q^2 + m_{FP}^2} dq d\theta, \quad (3.17b)$$

$v_e \approx 30 \text{ km s}^{-1}$ is the orbital velocity of earth. For direct comparison with the results in [20], we propose to plot the frequency shift as a function of time in days from the solar conjunction. At the solar conjunction, the impact parameter hit a minimum of $b_{min} = 1.6 R_\odot$ ($R_\odot$ is the Sun radius). Thus for a time t the impact parameter could be computed as $b(t) = v_e t + b_{min}$.

Figure 3.1 shows that the calculation of the gravitational frequency shift using the eikonal approximation within the model we proposed is in strong agreement with the overall behavior determined in the Cassini experiment. This is of high importance as it increases our confidence in the restriction derived on the theory parameters.

Using the fact that,

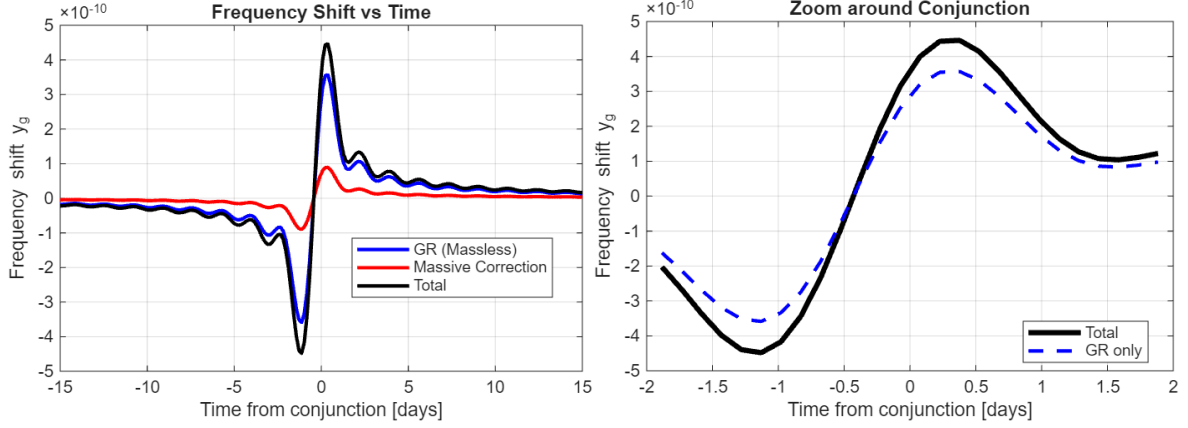
$$\int_0^{2\pi} e^{ibq \cos(\theta)} d\theta = 2\pi J_0(bq), \quad (3.18)$$

the ratio of the massive to the massless contribution to time delay is given by

$$r(\alpha, m_{FP}) = 4\alpha^4 \frac{\int_0^\Lambda \frac{J_0(bq)q}{q^2 + m_{FP}^2} dq}{\int_0^\Lambda \frac{J_0(bq)}{q} dq}. \quad (3.19)$$

Given that for theories beyond GR the time Shapiro time delay is expressed in terms of the post Newtonian parameter as

$$\Delta T = 2(1 + \gamma)\Delta T_{GR}, \quad (3.20)$$



(a) Frequency shift as a function of time delay. (b) Zoom on the solar conjunction peak

Figure 3.1: Frequency shift versus time delay. $\alpha = 0.5$, $m_{FP} = 10^{-40}$ eV, and $\Lambda = \frac{1}{b_{min}}$ eV. We get the same overall behavior as shown in Figure 2 [20], with the peak of $y_{max} = 6 \cdot 10^{-10}$ around the solar conjunction. The apparent oscillations are due to the Fourier truncation, the period is on the order $T = \frac{2\pi}{\Lambda v_e}$, thus for bigger cut-off the oscillation are faster. It is possible to surpass this oscillation by introducing a filter during the integration i.e. by adding a term of the form $e^{(-q/\Lambda)^n}$ where n is an integer. However because we are interested in the overall, and that the constraint derived later is based on the minimum impact parameter, we do not use this method.

the ratio $r(\alpha, m_{FP})$ us given by

$$r(\alpha, m_{FP}) = \frac{\gamma - 1}{2}. \quad (3.21)$$

The restriction from the Cassini space mission are obtained for $b = 1.6R_\odot$ where $R_\odot$ is the solar radius. We also express Λ in terms of the physical distances r_E the distance between earth and Sun and r_S distance between spacecraft and Sun $\Lambda = \frac{1}{\sqrt{r_E r_S}}$. The physical parameter space permitted is therefore

$$2r(\alpha, m_{FP}) \leq 2.1 \cdot 10^{-5}. \quad (3.22)$$

Figure 3.2 is a color map of the ratio r of the derivative of the massive to the massless contribution of time delay. The restriction from Cassini are also plotted. For small values of m_{FP} less than 10^{-30} [ev] the constraint is only dependent on the ratio of the Planck masses α . We find that for the theory to stay within the allowed range by the experiment α should be less than 0.06.

The behavior of constraint could be understood by studying the limits $\frac{m_{FP}}{\Lambda} \rightarrow 0$ and $\frac{m_{FP}}{\Lambda} \rightarrow +\infty$. First, we take the ultraviolet cut-off to be $\Lambda = \frac{1}{b_{min}}$, where

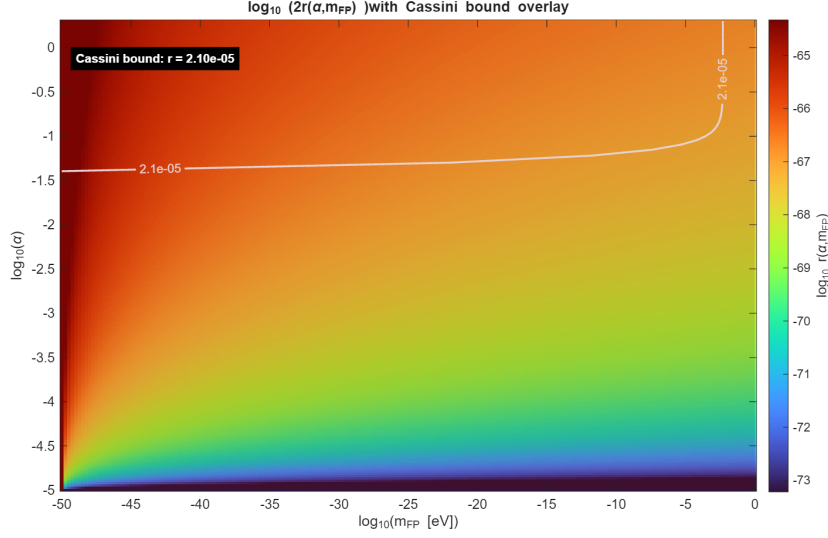


Figure 3.2: Color map for $r(\alpha, m_{FP})$. The map is in log scale. m_{FP} is taken in a range between 10^{-50} and 1 [ev]. α varying from 10^{-5} to 2. We also overlay the the constraint on the deviation from GR prediction. For small values of the Fierz-Pauli mass the constraint is mainly on α . The maximum allowed value of α is $10^{-1.219} \approx 0.06$

$b_{min} = 1.6R_{\odot} \approx$. In this case, letting $x = bq$ we can rewrite the ratio function as

$$r(\alpha, m_{FP}) = 4\alpha^4 \frac{\int_0^1 \frac{J_0(x)x}{x^2 + (\frac{m_{FP}}{\Lambda})^2} dx}{\int_0^1 \frac{J_0(x)}{x} dx}. \quad (3.23)$$

In the limit $\frac{m_{FP}}{\Lambda} \rightarrow 0$ we get

$$r(\alpha, m_{FP}) \approx 4\alpha^4 \quad (3.24)$$

Thus the constraint is independent of the mass term, explaining the plateau in figure 3.2. In the limit $\frac{m_{FP}}{\Lambda} \rightarrow +\infty$, we get

$$r(\alpha, m_{FP}) = 4\alpha^4 \frac{\Lambda^2}{m_{FP}^2} \frac{\int_0^1 J_0(x)x dx}{\int_0^1 \frac{J_0(x)}{x} dx}. \quad (3.25)$$

Then using the fact that $J_0(x)$ is positive over the range $[0, 1]$

$$\int_0^1 x J_0(x) dx \leq \int_0^1 J_0(x) dx, \quad (3.26a)$$

$$\int_0^1 \frac{J_0(x)}{x} dx \geq \int_0^1 J_0(x) dx. \quad (3.26b)$$

Which then lead to

$$r(\alpha, m_{FP}) \leq 4\alpha^4 \frac{\Lambda^2}{m_{FP}^2}. \quad (3.27)$$

Thus, when $\frac{m_{FP}}{\Lambda} \rightarrow +\infty$, $r(\alpha, m_{FP})$ will go to zero, independently of α . Hence, we can see why for large masses there is practically no restriction.

In [28], the author has calculated the post Newtonian parameter in HRBG to be

$$\gamma = \frac{3 + 2\alpha^2 e^{-m_{FP}r}}{3 + 4\alpha^2 e^{-m_{FP}r}}. \quad (3.28)$$

Using this calculation the authors in [29] have plotted the constraints on the the Fierz Pauli mass and the ratio of the Planck masses α . This can be see in Figure 6 in [29].

Hence forth we propose to compare our results to those from the classical calculations for different impact parameters, Fierz Pauli mass regimes and ratio of the Planck masses. We use the subscript 'cla' for entities derived from the post Newtonian expansion in [28] and 'eik' to those derived from the eikonal approximation. To explicitly compare our results to those obtained in [28], we define $r_{cla}(\alpha, m_{FP}) = \frac{\gamma_{cla}-1}{2}$ to be the equivalent to the ratio function derived earlier 3.25.

$$r_{cla}(\alpha, m_{FP}) = \frac{-\alpha^2 e^{-\frac{m_{FP}}{\Lambda}}}{3 + 4\alpha^2 e^{-\frac{m_{FP}}{\Lambda}}}. \quad (3.29)$$

In the small mass limit we have

$$r_{cla}(\alpha, m_{FP}) \approx \frac{-\alpha^2}{3 + 4\alpha^2}. \quad (3.30)$$

Thus if we are to also consider small values of α the ratio function scale quadratically, which is different for the power 4 scaling found in the eikonal approximation.

Figure 3.3 is a color map showing the relative difference between the post Newtonian parameter calculated in [28] and that in which we calculate using the eikonal approximation for different masses, impact parameters and Planck mass ratios. We also visualize the Vainshtein radius, the distance after which non linear effect become important and our first order approximation fails.

For small Fierz Pauli masses (less than 10^{-30} [ev]), the difference between the eikonal and classical calculation is relevant. However, as the mass gets bigger the discrepancy is only relevant for impact parameters smaller than the Vainshtein radius. This indicates that in the domain of validity of the first order expansion, beyond small mass regions, the eikonal calculations is successful in capturing the classical behavior.

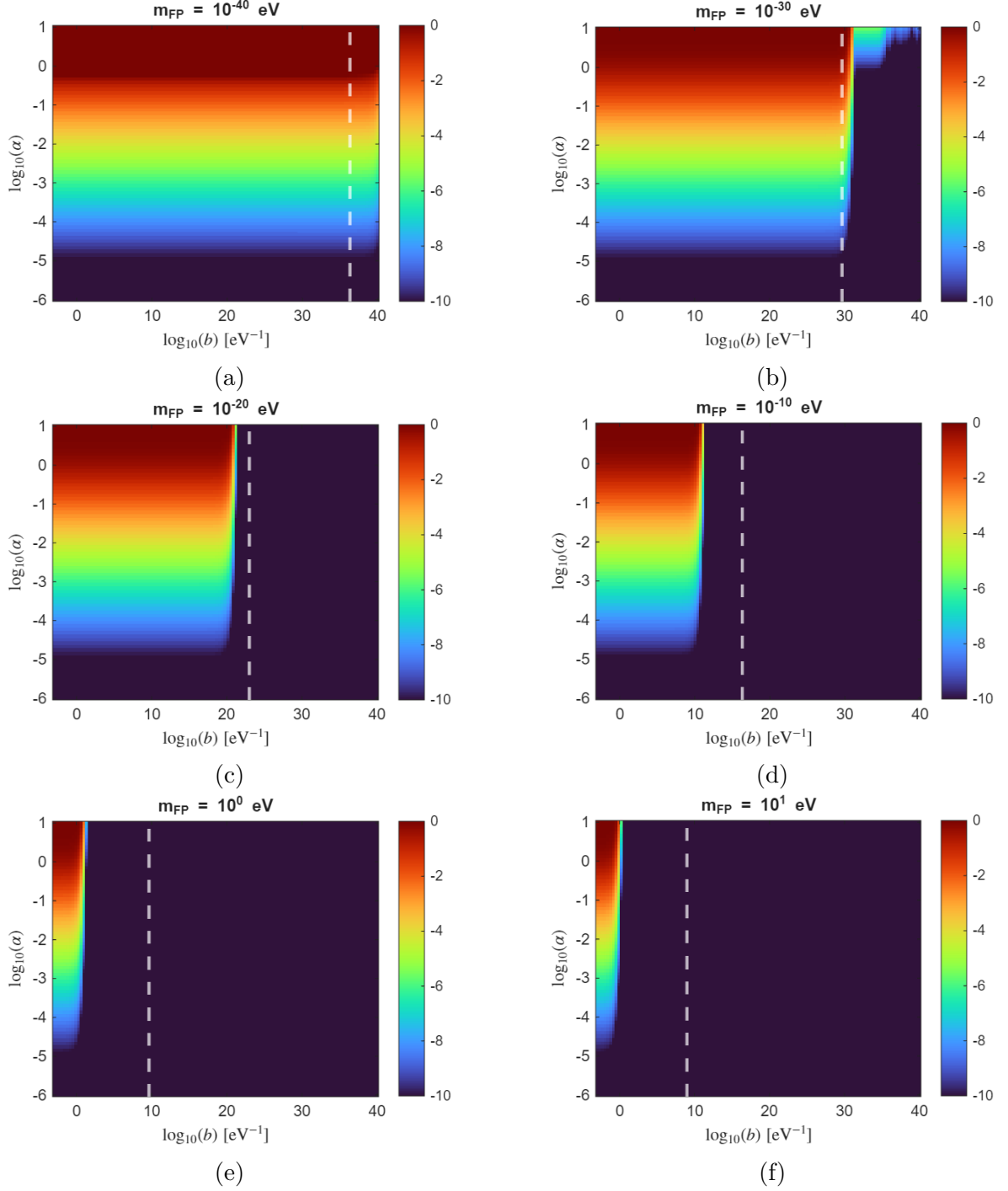


Figure 3.3: Relative difference between classical and eikonal PPN parameter γ for different Fierz–Pauli masses. The white dashed line indicates the Vainshtein radius, the distance after which the non linear effects are important and hence why the first order treatment fails. We observe that if the Fierz Pauli mass is larger than 10^{-30} eV then the difference between the classical and eikonal calculation is limited for distances smaller than the Vainshtein radius.

3.3 Chapter Conclusion

In this chapter, we applied the eikonal formalism developed previously to the solar-system regime and derived the gravitational time delay produced by the interaction of a photon with the Sun in bimetric gravity. After adapting the formalism to account for a massive scalar source and a massless probe, we computed the corresponding frequency shift, which is the observable directly measured by the Cassini spacecraft. The resulting time-dependent signal reproduces the key qualitative and quantitative features reported in [20], including the sharp peak around superior conjunction and the correct overall magnitude of the frequency shift. This agreement provides an important validation of the eikonal approach in the weak-field, large-impact-parameter regime relevant for solar-system tests.

By comparing our theoretical prediction with the experimental bound

$$\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5},$$

we extracted constraints on the bimetric parameters. In particular, we found that for small Fierz–Pauli masses $m_{\text{FP}} \lesssim 10^{-30}$ eV, the dominant restriction arises from the ratio of Planck masses, leading to the upper bound

$$\alpha \lesssim 0.06.$$

This result is consistent with, and complementary to, existing constraints derived from the PPN formalism, Vainshtein-screening analyses, and cosmological fits. All these independent approaches favour small values of α , and our calculation confirms this trend using a distinct scattering-based method. This strengthens the case for viable bimetric models with a suppressed coupling between the massive spin-2 mode and ordinary matter.

Overall, this chapter demonstrates that high-precision solar-system measurements impose strong constraints on bimetric gravity and that the eikonal framework provides a novel and reliable bridge between quantum-scattering techniques and classical gravitational observables. In the next chapter, we extend these ideas by applying the Wigner–Smith operator to the problem of gravitational time delay, exploring whether higher-particle intermediate states can reveal new features beyond the leading eikonal approximation.

Chapter 4

Application of Wigner Smith Operator to bimetric gravity

In the previous chapters, the time delay was computed by first transforming the scattering matrix element to the time domain and then applying the stationary phase approximation. This was straightforward in the context of the eikonal approximation because the scattering matrix element was already in an exponential form. However, if we wish to study systems where the momentum transfer is not negligible then the procedure fails because we do not expect the scattering matrix element to have such a convenient form. Thus, to be able to compute the time delay beyond the semi-classical regime we need a method that is robust and not approximation dependent. In this chapter we propose the use of the Wigner Smith operator as a tool to compute the time delay in an operator-based formalism. We will start by proving that in the eikonal regime, this operator reproduces the result obtained by applying the stationary phase approximation.

4.1 The Wigner Smith operator

4.1.1 Historical overview

In the study of scattering system, one observable that is hard to define is the interaction time.

A first answer to this problem was proposed by Eisenbud who suggested that the time delay a system experiences due to an interaction is encoded in the derivative of the phase shift. Later, Wigner [30], strengthened Eisenbud interpretation by deriving a

lower limit on the interaction time if one supposes that the interaction region is located within a sphere. One can see how this is problematic as the interaction radius is an arbitrarily defined quantity. In 1960 Smith [31] solved this problem, by proposing a more universal definition. Smith considered the difference in the propagation time a plane wave experiences with and without interaction. First, Smith defined the time a wave spends inside a sphere of radius R as the ratio of the probability of finding the particle inside the sphere and the probability flux in the sphere. The time delay for a radius R is then the difference between this quantity in the interacting case and in the free propagation case. The time delay the particle experience is the limit of this difference as the radius of the sphere approach infinity. Smith argued that if the interaction potential vanishes fast enough the limit converges and the time delay is well defined. Finally, when the limit exists, the time delay experienced by a state could be seen as the expectation value of a certain operator. This operator, later named WS operator, was then related to the scattering matrix of the system via [31]

$$Q = -iS^\dagger \frac{\partial S}{\partial E}. \quad (4.1)$$

where E is the energy of the system. Later on, the convergence of the WS operator was studied in the case of relativistic quantum mechanics by replacing the Schrodinger operator with the Klein Gordon and the Dirac operator. [32] presents a comprehensive review of time delay calculation in quantum mechanics.

While the WS operator was introduced in quantum mechanics, it is now mostly used in chaotic transport theory and electromagnetism [33]. A generalized form of the operator was introduced by replacing the derivative with respect to energy to other variables, such as position of scatter in a propagation region. In [34], the authors use this generalized WS operator to generate waves that focus on a specific scatter, resulting in a force.

4.1.2 Application of The Wigner Smith operator for scalar particle scattering in bimetric gravity

In this section, we will continue working in the eikonal regime given by the power scale 2.53. The goal is to study the behavior of the operator in this regime where we already have computed the time delay. Thus, gaining more confidence in the applicability of the WS before tackling regimes with large momentum transfer

We propose to calculate the WS operator matrix element between two massless scalars

incoming and outgoing state. The first obstacle we face when trying to extend this quantum mechanical operator to relativistic settings is the fact that, in its usual form, the operator is not relativistically invariant. This is due to the dependence of the energy of a system on the frame in which it is calculated. This problem is easily solved by using the relativistically invariant parameter $\sqrt{s}$ instead of the energy, hence

$$\langle p_3, p_4 | Q | p_1, p_2 \rangle = -i \langle p_3, p_4 | S^\dagger \frac{\partial S}{\partial \sqrt{s}} | p_1, p_2 \rangle. \quad (4.2)$$

To compute this matrix element we use the completeness relation inserting a complete set of intermediate states between the adjoint of the scattering matrix and its derivative. Given the initial state of two scalar particles of different types and the possible interaction vertices 2.1, the only relevant intermediate states are those where two massless scalars of different types are present with a given number of massive and massless gravitons. Thus we get,

$$\begin{aligned} \langle p_3, p_4 | Q | p_1, p_2 \rangle &= -i \sum_{n=0}^{\infty} \sum_{m=0}^M \sum_{\lambda_1 \dots \lambda_n} \sum_{\alpha_1 \dots \alpha_n} \int_{p'} \int_{p''} \prod_{i=1}^n \prod_{j=1}^m \int_{k^{\lambda_i}} \int_{l^{\alpha_j}} \\ &\times \langle p_3, p_4 | S^\dagger | p', p'', k^{\lambda_1}, \dots, k^{\lambda_n}, l^{\alpha_1}, \dots, l^{\alpha_m} \rangle \\ &\langle p', p'', k^{\lambda_1}, \dots, k^{\lambda_n}, l^{\alpha_1}, \dots, l^{\alpha_m} | \frac{\partial S}{\partial \sqrt{s}} | p_1, p_2 \rangle. \end{aligned} \quad (4.3)$$

In this formula the 2 initial sums are over the number of gravitons in the intermediate states. For the massless case (sum over n) the sum goes to infinity because the energy of the emitted massless gravitons can be arbitrary small, however for the massive case the number of massive gravitons allowed depend on the center of mass energy of the incoming state and on the mass of the graviton. The second and third sum are over the polarizations of the gravitons appearing in the intermediate state. Finally in the previous equation we used

$$\int_p := \int \frac{d^4 p}{(2\pi)^4} 2\pi \delta(p^2 + m^2) \Theta(p^0). \quad (4.4)$$

A more useful way to think about this equation is

$$\langle p_3, p_4 | Q | p_1, p_2 \rangle = \langle p_3, p_4 | Q_{2pc} | p_1, p_2 \rangle + \langle p_3, p_4 | Q_{3pc} | p_1, p_2 \rangle + \dots, \quad (4.5)$$

where Q_{2pc} is the two particle cut-off of the equation 4.3. In this work we will focus on the first two terms in this sum.

4.2 Two particle cut-off of the Wigner Smith operator

In this subsection we will focus on the two particle cut-off for the WS operator. We will show that the first term in the expansion 4.5 reproduces the time delay calculated using stationary phase approximation. To this end, we start by recalling the full two particle cut-off,

$$\langle p_3, p_4 | Q_{2pc} | p_1, p_2 \rangle = -i \int_{p'} \int_{p''} \langle p_3, p_4 | S^\dagger | p', p'' \rangle \frac{\partial}{\partial \sqrt{s}} \langle p', p'' | S | p_1, p_2 \rangle. \quad (4.6)$$

Recalling that 1.6

$$2\pi\delta(p_3^2)\Theta(p_3^0)2\pi\delta(p_4^2)\Theta(p_4^0) \langle p_3, p_4 | Q_{2pc} | p_1, p_2 \rangle = 2\pi^4\delta^{(4)}(p_1 + p_2 - p_3 - p_4)\hat{Q}_{2pc}(s, q), \quad (4.7)$$

we get

$$2\pi^4\delta^{(4)}(p_1 + p_2 - p_3 - p_4)\hat{Q}_{2pc}(s, q) = -i \int \frac{d^4 p'}{(2\pi)^4} \int \frac{d^4 p''}{(2\pi)^4} 2\pi\delta(p_3^2)\Theta(p_3^0)2\pi\delta(p_4^2)\Theta(p_4^0) \langle p_3, p_4 | S^\dagger | p', p'' \rangle 2\pi\delta(p'^2)\Theta(p'^0)2\pi\delta(p''^2)\Theta(p''^0) \frac{\partial}{\partial \sqrt{s}} \langle p', p'' | S | p_1, p_2 \rangle. \quad (4.8)$$

To advance further we propose the following definitions

$$q_1 = p_1 - p_3 = q, \quad q_2 = p_2 - p_4 = -q, \quad (4.9a)$$

$$q'_1 = p_1 - p', \quad q'_2 = p_2 - p'', \quad (4.9b)$$

$$q''_1 = p_3 - p' = q'_1 - q_1, \quad q''_2 = p_4 - p'' = q'_2 - q_2. \quad (4.9c)$$

Changing the integration to q'_1 and q'_2 , we can rewrite 4.8 as

$$2\pi^4\delta^{(4)}(q_1 + q_2)\hat{Q}_{2pc}(s, q) = -i \int \frac{d^4 q'_1}{(2\pi)^4} \int \frac{d^4 q'_2}{(2\pi)^4} 2\pi^4\delta^{(4)}(q'_1 - q_1 + q'_2 - q_2) \hat{S}^\dagger(s, q'_1 - q_1) 2\pi^4\delta^{(4)}(q'_1 + q'_2) \frac{\partial}{\partial \sqrt{s}} \hat{S}(s, q'_1) \quad (4.10)$$

Where we have again used the definition of the hatted functions in equation 1.6. Using $\delta^{(4)}(q'_1 + q'_2)$ to compute the integral over q''_2 , we get

$$\hat{Q}_{2pc}(s, q) = -i \int \frac{d^4 q'_1}{(2\pi)^4} \hat{S}^\dagger(s, q'_1 - q) \frac{\partial}{\partial \sqrt{s}} \hat{S}(s, q'_1), \quad (4.11a)$$

$$= -i(\hat{S}^\dagger \star_q \frac{\partial}{\partial \sqrt{s}} \hat{S})(s, q). \quad (4.11b)$$

$\star_q$ is the convolution operation over the q variables of 2 functions. Finally, transforming into the impact parameter space we get ¹

$$\tilde{\hat{Q}}_{2pc}(s, b) = -i \tilde{\hat{S}}^\dagger(s, b) \frac{\partial}{\partial \sqrt{s}} \tilde{\hat{S}}(s, b). \quad (4.12)$$

In the eikonal regime, the scattering matrix is given by

$$\tilde{\hat{S}}(s, b) = e^{2i\delta(s, b)}. \quad (4.13)$$

Using this and the fact that $(\tilde{\hat{S}}^\dagger) = (\tilde{\hat{S}})^\dagger$, we get

$$\hat{Q}_{2pc}(s, q) = \frac{\partial(2\delta(s, b))}{\partial \sqrt{s}}. \quad (4.14)$$

Thus, we see that the WS operator restricted to two-particle intermediates states reproduces the time delay calculated from the stationary phase approximation. The importance of this result is that it increases our confidence in the applicability of the WS operator to relativistic settings and the interpretation of the matrix element as a time delay experienced by the interacting system. However, if we recover the time delay already from the two particle cut-off the question remains regarding the meaning of the higher order terms in the expansion of the WS operator. To answer this question we propose to calculate the 3 particle cut-off contribution in the eikonal regime and see what physical information they hold.

4.3 The Three particle cut-off Wigner Smith operator with massless graviton intermediate state

In this section we focus on the three particles cut-off intermediate states. There are two contribution to this matrix element, the first one comes from an intermediate state

¹The derivative with respect to center of mass energy commute with the Fourier transform, because we suppose the the system energy and the momentum transfer are independent.

with a massless gravitons and the second one from a massive graviton. We will focus on the first contribution and we will work in the approximation that the energy of the intermediate graviton is smaller than that of the incoming and outgoing massless scalars. While this approximation simplifies the calculations, how valid it is remains questionable: The calculation of the three particle cut-off WS operator requires the integration over all possible momenta combination of the intermediate particles that respect energy momenta conservation. Hence, intermediate states where the energy of graviton is bigger than that of the intermediate boson should be included in the calculation of the matrix element. We note that due to energy momentum conservation, the graviton momenta is of the same order as the variation in the momenta of the two scalars. If one considers intermediate states that respect the eikonal power scaling, and where the angle of emission of the graviton is not too small (see next section for more details), the soft limit is achieved.

4.3.1 Inelastic scattering amplitude in impact parameter space

In the following, we are interested in the inelastic scattering process given by

$$\phi_1(p_1) + \phi_2(p_2) \rightarrow \phi'(p') + \phi''(p'') + k^\lambda(k).$$

p_1 and p_2 are the momenta of the two initial scalars, p' and p'' are the momenta of the final 2 scalars, and k is the momenta of the massless graviton. We also define $q = \frac{1}{2}(q_1 - q_2)$ where $q_1 = p_1 - p'$ and $q_2 = p_2 - p''$.

The eikonal regime is given by the following scaling hierarchy

$$|p_1 \cdot p_2| \gg |p_i \cdot q_j| \gg |q_i \cdot q_j|. \quad (4.15)$$

The energy momentum conservation translates to

$$q_1 + q_2 = k. \quad (4.16)$$

Taking the scalar product of this equation with p_1 we get

$$p_1 \cdot q_1 + p_1 \cdot q_2 = p_1 \cdot k, \quad (4.17a)$$

$$-p_1 \cdot k \ll -2p_1 \cdot p_2, \quad (4.17b)$$

$$p\omega(1 - \cos \theta) \ll s = 4p^2, \quad (4.17c)$$

$$\omega(1 - \cos \theta) \ll 4p = 2\sqrt{s}. \quad (4.17d)$$

ω is the graviton energy, p is the energy of the scalar ϕ_1 and θ is the graviton emission angle. We therefore observe that the eikonal regime entails the soft limit only if the graviton emission is not collinear. The soft limit is therefore will be supposed and during the integration a limiting angle, θ_{min} , will be introduced to avoid the breaking of the approximation.

The emission of the massless graviton could be seen as adding an external graviton line to a diagram of the $2 \rightarrow 2$ elastic process. Figure 4.1 shows the two different

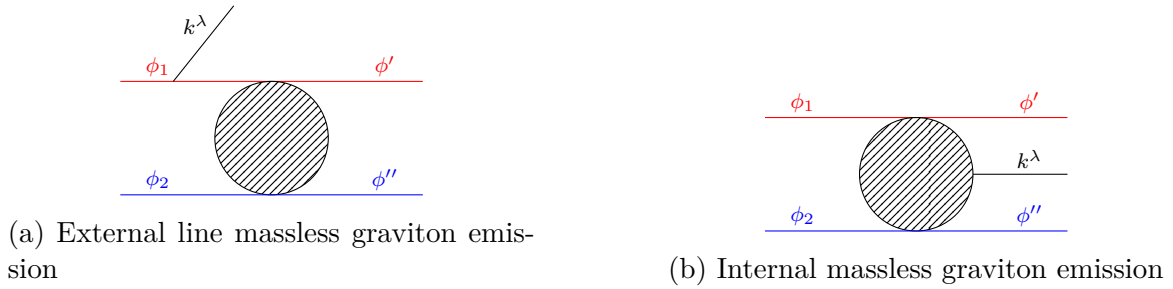


Figure 4.1: Starting from the elastic $2 \rightarrow 2$ scattering process, the inelastic $2 \rightarrow 3$ process with emission of massless graviton could be obtained either by an emission of the graviton by an external line 4.1a or by emission from an internal line 4.1b. The dashed circle represented the internal gravitons exchange and scalar propagators

ways a graviton can be added to the $2 \rightarrow 2$ process to obtain the inelastic process. Let $A_{2 \rightarrow 3}(s, q, k)$ be the amplitude for the inelastic process. In this amplitude we can factorize the polarization tensor of the emitted graviton, thus we can write

$$A_{2 \rightarrow 3}(s, q, k) = (\epsilon^\lambda)^\dagger_{\mu\nu} A_{2 \rightarrow 3}^{\mu\nu}(s, q, k). \quad (4.18)$$

Furthermore we can decomposes $A_{2 \rightarrow 3}^{\mu\nu}(s, q, k)$ into $A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k)$, the amplitude for emission from external scalar lines and $A_{(int)2 \rightarrow 3}^{\mu\nu}(s, q, k)$, the amplitude of the emission from internal scalar lines.

The amplitude for the process should be independent of the choice of the gauge in the quantization of the massless graviton field. This gauge invariance is given by the Ward identity

$$k_\mu A_{2 \rightarrow 3}^{\mu\nu}(s, q, k) = k_\nu A_{2 \rightarrow 3}^{\mu\nu}(s, q, k) = 0. \quad (4.19)$$

This identity is verified for the total amplitude i.e. $A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k)$ and $A_{(int)2 \rightarrow 3}^{\mu\nu}(s, q, k)$ do not necessarily obey this identity independently. This observation is important because, as we shall see later, in the soft limit the emission of the graviton from external

lines is leading. However, this by itself, is not enough to neglect the amplitude from internal lines emission, as we should also make sure that gauge freedom is respected to the leading order.

The fact that external graviton emission is leading for low energy graviton emission, could be seen as the result of the soft theorem derived by S. Weinberg in 1965 [35]. This theorem presents a central results in QFT and a similar result exist for photon emission in quantum electrodynamics. In this work, rather than stating the result of the theorem directly, we propose to perform the calculations.

Consider 4.1a as an example of external graviton emission, the amplitude for this diagram could be written as

$$A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) \approx G_1(p_1 - k) \tau^{(0)\mu\nu}(p_1, p_1 - k) A(s, q - \frac{k}{2}). \quad (4.20)$$

We can write the amplitude on this form because after the emission of the graviton the scattering proceeds as $2 \rightarrow 2$ process. Given that the graviton is soft, the center of mass energy and the momenta of the scalar ϕ_1 are only slightly changed. Thus, to get the $2 \rightarrow 3$ process we need to add the vertex for the graviton emission and the propagator of the energetic scalar. Furthermore the exchanged momentum between the 2 energetic scalars once the graviton is emitted is $q - \frac{k}{2}$ where $q = \frac{1}{2}(q_1 - q_2)$. In the soft regime the vertex coefficient is given by

$$\tau^{(0)\mu\nu}(p_1, p_1 - k) \approx 2i\gamma_G p_1^\mu p_1^\nu. \quad (4.21)$$

and the propagator is

$$G_1(p_1 - k) \approx \frac{i}{2p_1 \cdot k + i\epsilon}. \quad (4.22)$$

Consider next the emission from an internal line: The effect of the emission of massless

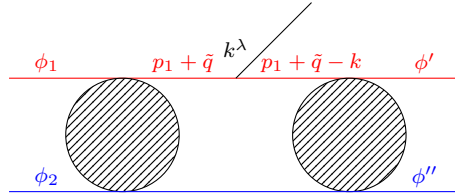


Figure 4.2: Graviton emission for an internal line. The effect of the emission is the addition of a massless vertex and a propagator to the elastic process

graviton from an internal line to the amplitude, is the addition of a vertex of a massless graviton and propagator. The vertex to first order will yield the same result as in the external emission case. The main difference is the addition of the propagator. Given

that we work in the soft limit we can Taylor expand the propagator.

$$G_1(p_1 + \tilde{q} - k) = G_1(p_1 + \tilde{q}) - \frac{\partial G_1}{\partial p^\mu}(p_1 + \tilde{q})k^\mu. \quad (4.23)$$

In the limit $k \rightarrow 0$ the propagator can safely be approximated with the zeroth order $G_1(p_1 + \tilde{q})$. Thus, we get

$$A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) \propto \frac{1}{k}, \quad (4.24a)$$

$$A_{(int)2 \rightarrow 3}^{\mu\nu}(s, q, k) \propto k^0. \quad (4.24b)$$

In the soft limit the external emission is therefore leading. Consider next the Ward identity 4.19. To leading order in k we obtain

$$k_\mu A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) = k_\nu A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) = 0. \quad (4.25)$$

Thus we can safely neglected the internal emission without violating the gauge symmetry. We now focus on the external graviton emission. The emission from the 4 external legs is given by

$$\begin{aligned} A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) = & -\gamma_G \left(\frac{p_1^\mu p_1^\nu}{p_1 \cdot k} - \frac{p'^\mu p'^\nu}{p' \cdot k} \right) A(s, q - \frac{k}{2}) - \\ & -\gamma_G \left(\frac{p_2^\mu p_2^\nu}{p_2 \cdot k} - \frac{p''^\mu p''^\nu}{p'' \cdot k} \right) A(s, q + \frac{k}{2}). \end{aligned} \quad (4.26)$$

We can further simplify this expression using $p' = p_1 - q_1 = p_1 - q - \frac{k}{2}$ and $p'' = p_2 - q_2 = p_2 + q - \frac{k}{2}$. Using gauge symmetry, terms that are proportional to k^μ could be eliminated. Thus we get

$$p'^\mu p'^\nu = p_1^\mu p_1^\nu - p_1^\mu q^\nu - p_1^\nu q^\mu, \quad (4.27a)$$

$$p''^\mu p''^\nu = p_2^\mu p_2^\nu + p_2^\mu q^\nu + p_2^\nu q^\mu. \quad (4.27b)$$

Furthermore, given that to get the full amplitude we need to contract with the polarization tensors of massless graviton which are symmetric, we can replace $p_1^\mu q^\nu + p_1^\nu q^\mu$ with $2p_1^\mu q^\nu$ and $p_2^\mu q^\nu + p_2^\nu q^\mu$ with $2p_2^\mu q^\nu$. Thus we get

$$A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q, k) = -2\gamma_g \left(\frac{p_1^\mu q^\nu}{p_1 \cdot k} A(s, q - \frac{k}{2}) - \frac{p_2^\mu q^\nu}{p_2 \cdot k} A(s, q + \frac{k}{2}) \right). \quad (4.28)$$

Transforming into the impact parameter space we get

$$\begin{aligned} \frac{1}{2s} \int \frac{d^2 q_\perp}{(2\pi)^2} A_{(ext)2 \rightarrow 3}^{\mu\nu}(s, q_\perp, k) e^{ib \cdot q_\perp} &= -2\gamma_G \frac{p_1^\mu}{p_1 \cdot k} \frac{1}{2s} \int \frac{d^2 q_\perp}{(2\pi)^2} q_\perp^\nu \\ &A\left(s, q_\perp - \frac{k}{2}\right) e^{ib \cdot q_\perp} + 2\gamma_G \frac{p_2^\mu}{p_2 \cdot k} \frac{1}{2s} \int \frac{d^2 q_\perp}{(2\pi)^2} q_\perp^\nu A\left(s, q_\perp + \frac{k}{2}\right) e^{ib \cdot q_\perp}. \end{aligned} \quad (4.29)$$

With the appropriate variable change for each integral we get

$$FT[A]_{(ext)2 \rightarrow 3}^{\mu\nu}(s, b, k) = -2\gamma_G \left(\frac{p_1^\mu}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\mu}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \frac{\partial}{\partial b_\nu} FT[A](s, b). \quad (4.30)$$

In obtaining 4.30, we have used gauge freedom to eliminate terms that are proportional to k^ν .

4.3.2 The three particle WS cut-off with massless intermediate state

With the transformation to impact parameter space of the soft massless graviton amplitude, we can finally determine the contribution of the massless graviton intermediate state to the three particle cut-off WS operator.²

$$\begin{aligned} \langle p_3, p_4 | Q_{3pc}^{(0)} | p_1 p_2 \rangle &= -i \sum_\lambda \int_k \int \frac{d^4 p'}{(2\pi)^4} \delta(p'^2) \int \frac{d^4 p''}{(2\pi)^4} \delta(p''^2) \langle p_3, p_4 | S^\dagger | p', p'', k^\lambda \rangle \\ &\frac{\partial}{\partial \sqrt{s}} \langle p', p'', k^\lambda | S | p_1, p_2 \rangle. \end{aligned} \quad (4.31)$$

The next steps are notation heavy so we present all the different notation used here before moving on

$$q_1 = p_1 - p_3, \quad q_2 = p_2 - p_4, \quad (4.32a)$$

$$q'_1 = p_1 - p', \quad q'_2 = p_2 - p'', \quad q' = \frac{1}{2}(q'_1 - q'_2), \quad (4.32b)$$

$$q''_1 = p_3 - p' = q'_1 - q_1, \quad q''_2 = p_4 - p'' = q'_2 - q_2, \quad q'' = \frac{1}{2}(q''_1 - q''_2) = \frac{1}{2}(q_1 - q_2) - q'. \quad (4.32c)$$

Inserting the definition of the scattering matrix

$$\langle p', p'', k^\lambda | S | p_1, p_2 \rangle = i(2\pi)^4 \delta^{(4)}(q'_1 + q'_2 - k) A_{2 \rightarrow 3}(s, q', k), \quad (4.33)$$

²Given that we later will only consider intermediate states that respect the eikonal approximation, we will neglect the heavy sides functions (Θ)

we get

$$\begin{aligned}
\langle p_3, p_4 | Q_{3pc}^{(0)} | p_1 p_2 \rangle &= -i \sum_{\lambda} \int_k \int \frac{d^4 q'_1}{(2\pi)^4} (2\pi) \delta(2p_1 \cdot q'_1) \int \frac{d^4 q'_2}{(2\pi)^4} (2\pi) \delta(2p_2 \cdot q'_2) \\
&\quad (2\pi)^4 \delta^{(4)}(q'_1 - q_1 + q'_2 - q_2 - k) A_{2 \rightarrow 3}^{\dagger}(s, \frac{1}{2}(q_1 - q_2) - q', k) \\
&\quad (2\pi)^4 \delta^{(4)}(q'_1 + q'_2 - k) \frac{\partial}{\partial \sqrt{s}} A_{2 \rightarrow 3}(s, q', k).
\end{aligned} \tag{4.34}$$

Multiplying both sides by $2\pi\delta(p_3^2)2\pi\delta(p_4^2)$ we get

$$\begin{aligned}
(2\pi)^4 \delta^{(4)}(q_1 + q_2) \hat{Q}_{3pc}^0(s, q_1) &= -i(2\pi)^4 \delta^{(4)}(q_1 + q_2) 2\pi\delta(2p_1 \cdot q_1) 2\pi\delta(2p_2 \cdot q_1) \\
&\quad \sum_{\lambda} \int_k \int \frac{d^4 q'}{(2\pi)^4} (2\pi) \delta(2p_1 \cdot (q' + \frac{k}{2})) (2\pi) \delta(2p_2 \cdot (\frac{k}{2} - q')) \\
&\quad A_{2 \rightarrow 3}^{\dagger}(s, q_1 - q', k) \frac{\partial}{\partial \sqrt{s}} A_{2 \rightarrow 3}(s, q', k).
\end{aligned} \tag{4.35}$$

Finally, to advance further we use the following approximation ³

$$\delta(2p_1 \cdot (q' + \frac{k}{2})) \approx \delta(2p_1 \cdot q'), \tag{4.36a}$$

$$\delta(2p_2 \cdot (\frac{k}{2} - q')) \approx \delta(-2p_2 \cdot q'). \tag{4.36b}$$

Thus, we get

$$\begin{aligned}
\hat{Q}_{3pc}^0(s, q_1) &= -i2\pi\delta(2p_1 \cdot q_1) 2\pi\delta(2p_2 \cdot q_1) \sum_{\lambda} \int_k \int \frac{d^2 q'_{\perp}}{2s(2\pi)^2} A_{2 \rightarrow 3}^{\dagger}(s, q_1 - q'_{\perp}, k) \\
&\quad \frac{\partial}{\partial \sqrt{s}} A_{2 \rightarrow 3}(s, q'_{\perp}, k),
\end{aligned} \tag{4.37a}$$

$$= -i2\pi\delta(2p_1 \cdot q_1) 2\pi\delta(2p_2 \cdot q_1) \sum_{\lambda} \int_k \left(A_{2 \rightarrow 3}^{\dagger} \star_{q_{\perp}} \frac{\partial}{\partial \sqrt{s}} A_{2 \rightarrow 3} \right) (s, q_1, k). \tag{4.37b}$$

³This approximation should hold for very low gravitons energy

$\star_{q_\perp}$ is the convolution operation associated with the Fourier transform FT . Transforming into impact parameter space, we get

$$\tilde{Q}_{3pc}^0(s, b) = -i \sum_\lambda \int_k FT \left[A_{2 \rightarrow 3}^\dagger \star_{q_\perp} \frac{\partial}{\partial \sqrt{s}} A_{2 \rightarrow 3} \right], \quad (4.38a)$$

$$= -i \sum_\lambda \int_k FT[A^\dagger]_{2 \rightarrow 3}(s, b, k^\lambda) \frac{\partial}{\partial \sqrt{s}} FT[A]_{2 \rightarrow 3}(s, b, k^\lambda), \quad (4.38b)$$

$$= -i \int_k \sum_\lambda \epsilon_{\rho\sigma}^\lambda(k) \epsilon_{\mu\nu}^{\dagger\lambda} FT[A^{\dagger\rho\sigma}]_{2 \rightarrow 3}(s, b, k^\lambda) \frac{\partial}{\partial \sqrt{s}} FT[A^{\mu\nu}]_{2 \rightarrow 3}(s, b, k^\lambda). \quad (4.38c)$$

In the soft eikonal approximation, we have

$$FT[A^{\mu\nu}]_{2 \rightarrow 3}(s, b, k^\lambda) = FT[A]_{(ext)2 \rightarrow 3}^{\mu\nu}(s, b, k) = -2\gamma_G \left(\frac{p_1^\mu}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\mu}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \frac{\partial}{\partial b_\nu} FT[A](s, b). \quad (4.39)$$

Using $FT[A](s, b) = e^{2i\delta_0(s, b)} - 1$, we get

$$FT[A]_{(ext)2 \rightarrow 3}^{\mu\nu}(s, b, k) = -2i\gamma_G \left(\frac{p_1^\mu}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\mu}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \frac{\partial 2\delta_0(s, b)}{\partial b_\nu} e^{2i\delta_0(s, b)}. \quad (4.40)$$

thus,

$$\begin{aligned} \frac{\partial}{\partial \sqrt{s}} FT[A]_{(ext)2 \rightarrow 3}^{\mu\nu}(s, b, k) &= -2i\gamma_G \left(\frac{p_1^\mu}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\mu}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \\ &\quad \left(\frac{\partial 2\delta_0(s, b)}{\partial b_\nu \partial \sqrt{s}} + i \frac{\partial 2\delta_0(s, b)}{\partial b_\nu} \frac{\partial 2\delta_0(s, b)}{\partial \sqrt{s}} \right) e^{2i\delta_0(s, b)}. \end{aligned} \quad (4.41)$$

To make the calculation easier we define $I^\nu = \frac{\partial 2\delta_0(s, b)}{\partial b_\nu}$ ⁴ and we use $\tilde{Q}_{2pc}(s, b) = \Delta T = \frac{\partial 2\delta_0(s, b)}{\partial \sqrt{s}}$.

$$\frac{\partial}{\partial \sqrt{s}} FT[A]_{(ext)2 \rightarrow 3}^{\mu\nu}(s, b, k) = -2i\gamma_G \left(\frac{p_1^\mu}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\mu}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \left(\frac{\partial I^\nu}{\partial \sqrt{s}} + i I^\nu \Delta T \right) e^{2i\delta_0(s, b)}. \quad (4.42)$$

⁴In [22], the authors prove that I^ν is the classical impulse of the scattering of the particles

Focusing on the Fourier transform of the adjoint amplitude we have

$$FT[A^{\dagger\rho\sigma}]_{(ext)2\rightarrow 3}(s, b, k) = (FT[A^{\rho\sigma}]_{2\rightarrow 3})^\dagger(s, -b, k), \quad (4.43a)$$

$$= 2i\gamma_G \left(\frac{p_1^\rho}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\rho}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \frac{\partial 2\delta_0^\dagger(s, -b)}{\partial b_\sigma} e^{-2i\delta_0^\dagger(s, -b)}, \quad (4.43b)$$

$$= 2i\gamma_G \left(\frac{p_1^\rho}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\rho}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) \frac{\partial 2\delta_0(s, b)}{\partial b_\sigma} e^{-2i\delta_0(s, b)}, \quad (4.43c)$$

$$= 2i\gamma_G \left(\frac{p_1^\rho}{p_1 \cdot k} e^{ib \cdot \frac{k}{2}} - \frac{p_2^\rho}{p_2 \cdot k} e^{-ib \cdot \frac{k}{2}} \right) I^\sigma e^{-2i\delta_0(s, b)}. \quad (4.43d)$$

In the last line we have used $\delta_0(s, -b)^\dagger = \delta_0(s, b)$ as can be easily checked from equation 2.80. Using 4.42 and 4.43d, we get

$$FT[A^{\dagger\rho\sigma}]_{(ext)2\rightarrow 3}(s, b, q) \frac{\partial}{\partial \sqrt{s}} FT[A^{\mu\nu}](s, b, k) = 4\gamma_G^2 \left(\frac{p_1^\mu p_1^\rho}{(p_1 \cdot k)^2} e^{ib \cdot k} - \frac{p_1^\mu p_2^\rho + p_1^\rho p_2^\mu}{(p_1 \cdot k)(p_2 \cdot k)} + \frac{p_2^\mu p_2^\rho}{(p_2 \cdot k)^2} e^{-ib \cdot k} \right) \left(I^\sigma \frac{\partial I^\nu}{\partial \sqrt{s}} + iI^\sigma I^\nu \Delta T \right). \quad (4.44)$$

Finally, in the Deonder gauge, the polarization tensors verify

$$\sum_\lambda \epsilon_{\mu\nu}^{\dagger\lambda} \epsilon_{\rho\sigma}^\lambda = \frac{1}{2} (\eta_{\mu\rho} \eta_{\nu\sigma} + \eta_{\mu\sigma} \eta_{\nu\rho} - \eta_{\mu\nu} \eta_{\rho\sigma}). \quad (4.45)$$

Before calculating the integrant in 4.37b, we make the following observation

$$I^\nu = \frac{\partial 2\delta_0}{\partial b_\nu} = \frac{is}{m_g^2} \int q_\perp^\nu e^{ib \cdot q_\perp} \left(\frac{1}{4q_\perp^2} + \frac{\alpha^4}{q_\perp^2 + m_{FP}^2} \right) \frac{d^2 q_\perp}{(2\pi)^2}. \quad (4.46)$$

Thus,

$$p_i \cdot I = \frac{is}{m_g^2} \int (p_i \cdot q_\perp) e^{ib \cdot q_\perp} \left(\frac{1}{4q_\perp^2} + \frac{\alpha^4}{q_\perp^2 + m_{FP}^2} \right) \frac{d^2 q_\perp}{(2\pi)^2} = 0. \quad (4.47)$$

For $i = 1, 2$. Where we have used the definition of the perpendicular momentum transfer $q_\perp$. This simplifies the calculation as the only non zero term in the contraction

is the first. Hence we get

$$\begin{aligned} \sum_{\lambda} \epsilon_{\rho\sigma}^{\lambda}(k) \epsilon_{\mu\nu}^{\dagger\lambda} FT[A^{\dagger\rho\sigma}]_{2\rightarrow 3}(s, b, k^{\lambda}) \frac{\partial}{\partial \sqrt{s}} FT[A^{\mu\nu}]_{2\rightarrow 3}(s, b, k^{\lambda}) \\ = 2\gamma_G^2 \frac{-2(p_1 \cdot p_2)^2}{(p_1 \cdot k)(p_2 \cdot k)} \left(\frac{1}{2} \frac{\partial I^2}{\partial \sqrt{s}} + iI^2 \Delta T \right), \end{aligned} \quad (4.48a)$$

$$= -\frac{\gamma_G^2 s^2}{(p_1 \cdot k)(p_2 \cdot k)} \left(\frac{1}{2} \frac{\partial I^2}{\partial \sqrt{s}} + iI^2 \Delta T \right). \quad (4.48b)$$

Finally we get

$$\tilde{Q}_{3pc}^0(s, b) = i\gamma_G^2 s^2 \left(\frac{1}{2} \frac{\partial I^2}{\partial \sqrt{s}} + iI^2 \Delta T \right) \int \frac{d^4 k}{(2\pi)^4} 2\pi \delta(k^2) \Theta(k^0) \frac{1}{(p_1 \cdot k)(p_2 \cdot k)}. \quad (4.49)$$

To compute the integral appearing in 4.49, we suppose that the momenta of the particles ϕ_1 and ϕ_2 are along the z axis

$$p_1 = (p, 0, 0, p) \quad p_2 = (p, 0, 0, -p). \quad (4.50)$$

Hence we get

$$\tilde{Q}_{3pc}^0(s, b) = i\gamma_G^2 s^2 \left(\frac{1}{2} \frac{\partial I^2}{\partial \sqrt{s}} + iI^2 \Delta T \right) \int \frac{d^3 k}{(2\pi)^3 (2\omega)} \frac{1}{\omega}, \quad (4.51a)$$

$$= i\frac{\gamma_G^2 s}{2\pi^2} \left(\frac{1}{2} \frac{\partial I^2}{\partial \sqrt{s}} + iI^2 \Delta T \right) \int \frac{1}{\omega} d\omega \int_0^\pi \frac{1}{\sin(\theta)} d\theta. \quad (4.51b)$$

In the last line we have used $s = -(p_1 + p_2)^2 = 4p^2$. Before discussing the meaning of the this cut-offs, we note that $\frac{\partial I^2}{\partial \sqrt{s}} = \frac{4}{\sqrt{s}} I^2$ because $I^2 \propto s^2$. Thus we can rewrite 4.51 as

$$\tilde{Q}_{3pc}^0(s, b) = \frac{i\gamma_G^2}{2\pi^2} I^2 \left(2\sqrt{s} + is\tilde{Q}_{2pc}(s, b) \right) \int_{\Lambda_{IR}}^{\Lambda_{UV}} \frac{1}{\omega} d\omega \int_{\theta_{min}}^{\pi-\theta_{min}} \frac{1}{\sin(\theta)} d\theta, \quad (4.52a)$$

$$= -\frac{\gamma_G^2 s}{2\pi^2} I^2 \left(\tilde{Q}_{2pc}(s, b) - i\frac{2}{\sqrt{s}} \right) \ln \left(\frac{\Lambda_{UV}}{\Lambda_{IR}} \right) \left(-2 \ln \left(\tan \left(\frac{\theta_{min}}{2} \right) \right) \right). \quad (4.52b)$$

Λ_{IR} and Λ_{UV} are the infrared and ultraviolet cut-offs and θ_{min} is the minium allowed angle between the graviton and the energetic scalars momenta. The angular integral is calculated by using the variable change $u = \tan \left(\frac{\theta_{min}}{2} \right)$. The ultraviolet cut-off Λ_{UV} is expected given that the previous derivation was based on the hypotheses that the eikonal regime is valid and hence the graviton is soft. The infrared cut-off on the other hand is due to the long range nature of massless gravity. The angular part also presents a logarithmic divergent for small θ angle i.e. if the graviton is emitted and reabsorbed

in parallel to the energetic scalars. θ_{min} is introduced as an angular cut-off. At the start of this section, we argued that for the eikonal regime to imply the soft graviton limit the emission (reabsorption) angle of the graviton should not be too small. Hence, our results break for small angles and this domain is not covered by the derived equation.

In [22], the authors calculate the classical impulse amplitude to be

$$I(s, b) = \frac{\gamma_G^2 s}{2\pi b}. \quad (4.53)$$

Where b is the amplitude of the impact parameter 4 vector. Injecting this into the 3 particle cut of we get

$$\tilde{Q}_{3pc}^0(s, b) = \frac{\gamma_G^4 s^2}{2\pi^3 b^2} \left(\tilde{Q}_{2pc}(s, b) - i \frac{2}{\sqrt{s}} \right) \ln \left(\frac{\Lambda_{UV}}{\Lambda_{IR}} \right) \ln \left(\tan \left(\frac{\theta_{min}}{2} \right) \right). \quad (4.54)$$

We also observe that the three particle massless cut-off is proportional to the amplitude of the classical impulse squared. The real part of $\tilde{Q}_{3pc}^0(s, b)$ is proportional to $s^{\frac{5}{2}}$ and the imaginary part is proportional to $s^{\frac{3}{2}}$.

If the WS operator entails the time delay a system experience due to interaction, then the imaginary part of equation 4.54 presents a problem. In QFT the imaginary part of the amplitude is due to the intermediate states getting on shell and producing real particles. This process then decreases the probability of the elastic scattering. Thus the imaginary part is to be expected. The real part encodes the correction to the time delay calculated with the 2 particle cut-off,

$$\tilde{Q}_{2pc}(s, b) + Re \left(\tilde{Q}_{3pc}^0(s, b) \right) = \tilde{Q}_{2pc}(s, b) \left(1 + \frac{\gamma_G^4 s^2}{2\pi^3 b^2} \ln \left(\frac{\Lambda_{UV}}{\Lambda_{IR}} \right) \ln \left(\tan \left(\frac{\theta_{min}}{2} \right) \right) \right). \quad (4.55)$$

For realistic θ_{min} values i.e. $\theta_{min} < \frac{\pi}{2}$ we get $\ln \left(\tan \left(\frac{\theta_{min}}{2} \right) \right) < 0$, thus the contribution to the time delay is negative resulting in unreal time advance. However, this results should be taken with caution. First, a sign mistake is highly possible. Second, this just the 3 particle cut-off contribution in the soft limit and this could be corrected by summing higher order diagrams with more intermediate massless gravitons. Third, this could be the result of the accumulation of multiple approximation.

4.4 Chapter Conclusion

In this chapter we investigated the novel application of the WS operator in gravitational time delay calculations. Given an initial and a final state, we showed that the matrix element of the operator can be written as an infinite series over suitable intermediate states. We then showed that the first term in this sum reproduces the results obtained by applying the stationary phase approximation.

We then investigated the next term in the series, i.e. the contribution of three particles intermediate states. Out of these, we calculated the contribution of intermediate states with one massless graviton. Assuming that the emission of this graviton is not collinear, we showed that the eikonal approximation results in soft gravitons. Through the soft theorem, we calculated the amplitude for the inelastic process and the Fourier transform to the impact parameter space.

The three particle cut-off WS operator is calculated. We note that the expression we derived suffers from infra red and ultraviolet divergence.

Finally, we note that if the emission (absorption) graviton angle is small, then the three particle WS operator cut-off real part is negative. While this by itself is not enough to rule out our results, it does point to a possible error either in the calculation or in the conceptual application of the operator. It is important to mention that the final result was obtained after a series of approximations that are not all rigorous. Hence, it is possible that our approximations have broken at some stage.

Thus, while the WS operator present a promising extension beyond the eikonal regime, more work is needed to prove the convergence of the infinite sum over intermediate states and in interpreting each term.

Chapter 5

Discussion And Conclusion

5.1 Conclusion

The aim of this work was to calculate the time delay a system of interacting particles experiences due to gravitational interaction in HRBG.

In the second chapter, we linearized the theory around the convenient background of a proportional flat solution. This allowed us to identify two modes: a massless spin-two particle identical to that present in classical GR, and a massive spin-two particle. Such diagonalization of the theory allowed for the interpretation of gravitational interaction as the exchange of a massless and a massive spin-two particles. Having prepared the groundwork, we considered the simplest possible scenario in which two massless spin-zero particles interact only through gravity. The scattering amplitude was then computed in the eikonal regime. Finally, using the stationary phase method, the time delay was computed from the eikonal phase. We observed that there are two contributions to the time delay: one from the massless mode, producing the result of GR, and the other arising from the massive mode contribution to the interaction.

In the third chapter of this thesis, we explored how the results of the first chapter can be extended to the study of light-ray delay in the solar system. First, we adapted the framework from the first chapter to the case of a massive scalar field (the Sun) and a massless one (the photon). The time delay was then computed. Using the Cassini space mission constraints on the post-Newtonian parameter, we determined the region of parameter space where the theory is in accordance with the measurements. We concluded the chapter by comparing the results of our calculation of the post-Newtonian parameter with those found in the literature. We found that beyond the Vainshtein radius, where the linear approximation is valid, the eikonal results are in

strong agreement with those in the literature.

The fourth chapter of this thesis focused on the application of the WignerSmith operator in HRBG. The aim of this chapter was to provide a possible method to move beyond the stationary phase approximation. This is motivated by the fact that we do not expect to be able to express the scattering amplitude as a simple exponential once higher-order terms are included in the action of the theory. We introduced the WS operator and expressed its matrix element as a sum over intermediate states. We showed that the first of these terms reproduces the result of the eikonal approximation. Exploring the higher-order terms, we considered intermediate states where a massless graviton is emitted and then absorbed again. The result of this calculation is a complex term that suffers from collinear and infrared divergences in the energy of the graviton. This complex term can be traced back to the inelastic nature of the process.

The novel contribution of this work lies in extending eikonal techniques to bimetric gravity, thereby incorporating scattering processes involving massless scalar fields mediated by the exchange of a massive graviton. We further generalize the analysis to scattering between one massless and one massive scalar field, which allows us to exploit Solar System observations to derive phenomenological constraints. Finally, we explore the application of the ws operator within the framework of bimetric gravity.

5.2 Discussion

As with any scientific work, this thesis suffers from multiple limitations, and several points can indeed be questioned. Here, we summarize what we consider to be the most important limitations of this work.

The main shortcoming of this work is that it fails to address higher-order terms in the action, thus failing to probe the causal behavior of the theory beyond the linear regime, which was already expected to be safe. As a result, the question of whether the interaction between the massive and massless gravitons produces acausal behavior remains unanswered.

In the third chapter of this work, modeling the Sun as a massive scalar particle, while producing consistent results, is still a strong approximation, and its validity needs to be discussed further. The spin of the photon was neglected; however, this can be justified by noting that Shapiro time-delay calculations also do not take spin effects into account. In any case, studying the interaction of bimetric gravity fields with a more realistic model is worth pursuing. At the end of this chapter, the results of the

eikonal approximation were compared to those in the literature. While agreement was found at the level of numerical results, we failed to reproduce the analytical expression of the post-Newtonian regime. Thus, whether this agreement is merely due to the chosen values of the parameters m_{FP} and α , or whether it indicates that the eikonal calculations can indeed reproduce the results of classical direct calculations, remains undecided.

The application of the WS operator to bimetric gravity is not fully justified. While the convergence of this operator has been studied for Schrödinger, KleinGordon, and more restrictively for Dirac-type operators, it has not been studied in the case of the kinetic operators encountered in gravitational theories. Thus, how far one can go by applying it to problems in bimetric gravity remains unclear. Beyond this mathematical subtlety, in the calculation of the matrix element we only considered intermediate states with zero or one soft massless graviton. Intermediate states involving one massive graviton were not considered, which represents a shortcoming since a comparison is therefore not possible. Furthermore, the interpretation of the massless one-graviton intermediate states themselves, as well as the divergences they exhibit, is not entirely clear.

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Appendix A

Linearization of Bimetric Field Equations

Since many results in the main text (spectrum, coupling, time delay) rely on the linearized equations, we provide here a step-by-step derivation for completeness. Recalling that the 2 metrics in the Hassan-Rosen theory verify the equations

$$R_{\mu\nu}(g) - \frac{1}{2}g_{\mu\nu}R(g) + \frac{m^4}{m_g^2}V_{\mu\nu}^g(g, f, \beta_n) = 0, \quad (\text{A.1a})$$

$$R_{\mu\nu}(f) - \frac{1}{2}f_{\mu\nu}R(f) + \frac{m^4}{m_g^2}V_{\mu\nu}^f(g, f, \beta_n) = 0. \quad (\text{A.1b})$$

We propose to linearize the theory around a proportional background solutions, thus we consider

$$g_{\mu\nu} = \bar{g}_{\mu\nu} + \delta g_{\mu\nu}, \quad (\text{A.2a})$$

$$f_{\mu\nu} = \bar{f}_{\mu\nu} + \delta f_{\mu\nu}. \quad (\text{A.2b})$$

Where $\bar{g}_{\mu\nu}$ and $\bar{f}_{\mu\nu}$ are 2 proportional solutions for the bimetric theory field equations related through $\bar{f}_{\mu\nu} = c^2 \bar{g}_{\mu\nu}$. $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$ are 2 distinct perturbations of the fields. We focus on A.2a and we propose to derive the equation verified by the perturbation $\delta g_{\mu\nu}$. We assume that $g_{\mu\nu}$ is a solution to A.1a and that $\bar{g}_{\mu\nu}$ verifies the proportional background equation 2.13. We propose to achieve this goal in 2 steps first, we focus on the Einstein tensor part, then we consider the variation of the potential.

A.1 Variation of the Einstein Tensor

The starting point is the Christoffel symbols $\Gamma_{\mu\nu}^\rho$

$$\Gamma_{\mu\nu}^\rho = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \quad (\text{A.3})$$

Expanding the metric $g_{\mu\nu}$ in terms of the background and the perturbation and using the property $\bar{g}^{\rho\lambda}\delta g_{\lambda\sigma} = -\delta g^{\rho\lambda}\bar{g}_{\lambda\sigma}$, it is straightforward to prove

$$\Gamma_{\mu\nu}^\rho = \bar{\Gamma}_{\mu\nu}^\rho + \delta\Gamma_{\mu\nu}^\rho, \quad (\text{A.4a})$$

$$\delta\Gamma_{\mu\nu}^\rho = \frac{1}{2}\bar{g}^{\rho\sigma}(\nabla_\mu g_{\sigma\nu} + \nabla_\nu g_{\mu\sigma} - \nabla_\sigma g_{\mu\nu}). \quad (\text{A.4b})$$

where $\bar{\Gamma}_{\mu\nu}^\rho$ is the Christoffel symbol for the background metric. From here onward quantities with a bar on top are defined with respect to the background metric, except for the covariant derivatives that are always defined with respect to the background metric yet we don't use a bar to distinguish them as we don't use other covariant derivative. As we later take the covariant derivative of $\delta\Gamma_{\mu\nu}^\rho$, we recall that while neither $\bar{\Gamma}_{\mu\nu}^\rho$ nor $\Gamma_{\mu\nu}^\rho$ are tensors, their difference is. This could be seen directly from the form of the variation, or by directly considering the transformation properties of the difference of the Christoffel symbols. Next we consider the Riemann tensor

$$R_{\sigma\mu\nu}^\rho = \partial_\mu \Gamma_{\sigma\nu}^\rho - \partial_\nu \Gamma_{\sigma\mu}^\rho + \Gamma_{\mu\lambda}^\rho \Gamma_{\sigma\nu}^\lambda - \Gamma_{\nu\lambda}^\rho \Gamma_{\sigma\mu}^\lambda, \quad (\text{A.5a})$$

$$= \bar{R}_{\sigma\mu\nu}^\rho + \delta R_{\sigma\mu\nu}^\rho. \quad (\text{A.5b})$$

Where

$$\delta R_{\sigma\mu\nu}^\rho = \nabla_\mu \delta\Gamma_{\sigma\nu}^\rho - \nabla_\nu \delta\Gamma_{\sigma\mu}^\rho. \quad (\text{A.6})$$

Next, we focus on the first part of the Einstein tensor, the Ricci tensor, it is obtained from Riemann tensor by contracting the first and third indices. Thus we get

$$R_{\sigma\nu} = R_{\sigma\rho\nu}^\rho, \quad (\text{A.7a})$$

$$= \bar{R}_{\sigma\nu} + \delta R_{\sigma\nu}. \quad (\text{A.7b})$$

with

$$\delta R_{\mu\nu} = \nabla_\rho \delta\Gamma_{\mu\nu}^\rho - \nabla_\nu \delta\Gamma_{\mu\rho}^\rho, \quad (\text{A.8a})$$

$$= \frac{1}{2}(\delta_\nu^\rho \nabla^\sigma \nabla_\mu \delta g_{\rho\sigma} + \delta_\mu^\rho \nabla^\sigma \nabla_\nu \delta g_{\rho\sigma} - \delta_\mu^\rho \delta_\nu^\sigma \nabla^2 \delta g_{\rho\sigma} - \bar{g}^{\rho\sigma} \nabla_\mu \nabla_\nu \delta g_{\rho\sigma}). \quad (\text{A.8b})$$

The curvature scalar is obtained from the contraction of the Ricci tensor,

$$R = g^{\rho\sigma} R_{\rho\sigma}, \quad (\text{A.9a})$$

$$= \bar{R} + \delta R. \quad (\text{A.9b})$$

$$(\text{A.9c})$$

with

$$\delta R = \bar{g}^{\rho\sigma} \delta R_{\rho\sigma} + \delta g^{\rho\sigma} \bar{R}_{\rho\sigma}. \quad (\text{A.10})$$

The first part of the variation of the Ricci scalar yields

$$\bar{g}^{\rho\sigma} \delta R_{\rho\sigma} = \nabla^\rho \nabla^\sigma \delta g_{\rho\sigma} - \nabla^2 \delta g. \quad (\text{A.11})$$

with $\delta g = \bar{g}^{\rho\sigma} \delta g_{\rho\sigma}$. Using the background equations 2.13, we can rewrite the the background Ricci tensor in terms of the background curvature scalar and the cosmological constant.

$$\bar{R}^{\rho\sigma} = \frac{1}{2} \bar{R} \bar{g}^{\rho\sigma} - \Lambda_g \bar{g}^{\rho\sigma}. \quad (\text{A.12})$$

First, we rewrite $\delta g^{\rho\sigma} \bar{R}_{\rho\sigma} = -\delta g_{\rho\sigma} \bar{R}^{\rho\sigma}$, here we used $\bar{g}^{\rho\lambda} \delta g_{\lambda\sigma} = -\delta g^{\rho\lambda} \bar{g}_{\lambda\sigma}$. Thus we get

$$\delta g^{\rho\sigma} \bar{R}_{\rho\sigma} = -\delta g_{\rho\sigma} \left(\frac{1}{2} \bar{R} \bar{g}^{\rho\sigma} - \Lambda_g \bar{g}^{\rho\sigma} \right), \quad (\text{A.13a})$$

$$= -\frac{1}{2} \bar{R} \delta g + \Lambda_g \delta g. \quad (\text{A.13b})$$

Taking the trace of the background equation $\bar{G}_{\mu\nu} + \Lambda_g \bar{g}_{\mu\nu} = 0$, we get $\bar{R} = 4\Lambda_g$. Injecting this in the previous equation we get

$$\delta g^{\rho\sigma} \bar{R}_{\rho\sigma} = -\Lambda_g \delta g. \quad (\text{A.14})$$

Now, we can consider the variation of the Einstein tensor:

$$\delta G_{\mu\nu} = \delta R_{\mu\nu} - \frac{1}{2} \bar{R} \delta g_{\mu\nu} - \frac{1}{2} \delta R \bar{g}_{\mu\nu}, \quad (\text{A.15a})$$

$$= \delta R_{\mu\nu} - \frac{1}{2} \bar{g}^{\rho\sigma} \delta R_{\rho\sigma} \bar{g}_{\mu\nu} - 2\Lambda_g \delta g_{\mu\nu} + \frac{1}{2} \Lambda_g \delta g \bar{g}_{\mu\nu}, \quad (\text{A.15b})$$

$$= \varepsilon_{\mu\nu}{}^{\sigma\rho} \delta g_{\rho\sigma} - 2\Lambda_g \delta g_{\mu\nu} + \frac{1}{2} \Lambda_g \delta g \bar{g}_{\mu\nu}. \quad (\text{A.15c})$$

With

$$\varepsilon_{\mu\nu}{}^{\rho\sigma} = -\frac{1}{2} \left[\delta_\mu^\rho \delta_\nu^\sigma \nabla^2 + \bar{g}^{\rho\sigma} \nabla_\mu \nabla_\nu - \delta_\mu^\rho \nabla^\sigma \nabla_\nu - \delta_\nu^\sigma \nabla^\rho \nabla_\mu - \bar{g}_{\mu\nu} \bar{g}^{\rho\sigma} \nabla^2 + \bar{g}_{\mu\nu} \nabla^\rho \nabla^\sigma \right]. \quad (\text{A.16})$$

A.2 Variation of the potential

After deriving the expression for the variation of the Einstein tensor, we now focus in the variation of the potential.

$$V_{\mu\nu}^g(g, f, \beta_n) = g_{\mu\rho} \sum_{n=0}^3 (-1)^n \beta_n (Y_n)^\rho{}_\nu(S). \quad (\text{A.17})$$

We start by expressing the variation of the matrix S in terms of the perturbation $\delta g_{\mu\nu}$ and $\delta f_{\mu\nu}$.

$$S^2{}^\rho{}_\nu = (\bar{S} + \delta S)^2{}^\rho{}_\nu, \quad (\text{A.18a})$$

$$= \bar{S}^2{}^\rho{}_\nu + 2\bar{S}^{\rho\sigma} \delta S_{\sigma\nu}. \quad (\text{A.18b})$$

Where we only consider terms to first order in the perturbation. Using the definition of S matrix in terms of the metrics $g_{\mu\nu}$ and $f_{\mu\nu}$, we get

$$S^2{}^\rho{}_\nu = (\bar{g}^{\rho\sigma} + \delta g^{\rho\sigma})(\bar{f}_{\sigma\nu} + \delta f_{\sigma\nu}), \quad (\text{A.19a})$$

$$= \bar{S}^2 + \bar{g}^{\rho\sigma} \delta f_{\sigma\nu} + \delta g^{\rho\sigma} \bar{f}_{\sigma\nu}, \quad (\text{A.19b})$$

$$= \bar{S}^2 + \bar{g}^{\rho\sigma} \delta f_{\sigma\nu} - c^2 \bar{g}^{\rho\sigma} \delta g_{\sigma\nu}. \quad (\text{A.19c})$$

Comparing equations A.18 and A.19, we get

$$2\bar{S}^{\rho\sigma} \delta S_{\sigma\nu} = \bar{g}^{\rho\sigma} \delta f_{\sigma\nu} - c^2 \bar{g}^{\rho\sigma} \delta g_{\sigma\nu}, \quad (\text{A.20a})$$

$$\delta S^\rho{}_\nu = \frac{1}{2c} \bar{g}^{\rho\sigma} (\delta f_{\sigma\nu} - c^2 \delta g_{\sigma\nu}). \quad (\text{A.20b})$$

Next we consider the variation of the matrices $(Y_n)^\rho{}_\sigma(S)$ appearing in the expression of the potential

$$(Y_n)^\rho{}_\sigma(S) = \sum_{k=0}^n (-1)^k e_k(\bar{S} + \delta S) (\bar{S} + \delta S)^{n-k}{}^\rho{}_\nu, \quad (\text{A.21a})$$

$$= \sum_{k=0}^n (-1)^k (e_k(\bar{S}) + \delta e_k) (\bar{S}^{n-k} + (n-k) \bar{S}^{n-k-1} \delta S)^\rho{}_\nu, \quad (\text{A.21b})$$

$$= (Y_n)^\rho{}_\sigma(\bar{S}) + \sum_{k=1}^n (-1)^k \delta e_k (\bar{S}^{n-k})^\rho{}_\nu + \sum_{k=0}^{n-1} (-1)^k e_k(\bar{S}) (n-k) (\bar{S}^{n-k-1} \delta S)^\rho{}_\nu. \quad (\text{A.21c})$$

The first sum starts from 0 because $\delta e_0 = 0$ and the second sum terminates at $n-1$ because the S in the n th term of the sum is to the power zero, and thus there is no variation for this terms. We focus first on the second term in the variation, using the fact that $\bar{S}^\rho{}_\nu = c \delta^\rho{}_\nu$. We get

$$\sum_{k=0}^{n-1} (-1)^k e_k(\bar{S}) (n-k) (\bar{S}^{n-k-1} \delta S)^\rho{}_\nu = \sum_{k=0}^{n-1} (-1)^k (n-k) e_k(\bar{S}) c^{n-k-1} (\delta S)^\rho{}_\nu. \quad (\text{A.22})$$

Next, we focus on the first term in the variation. Using the expression for the variation of the symmetric elementary polynomials 2.4.

$$\delta e_k(\bar{S}) = - \sum_{j=1}^k (-1)^j \text{Tr}(\bar{S}^{j-1} \delta S) e_{k-j}(\bar{S}), \quad (\text{A.23a})$$

$$= \sum_{j=1}^k (-1)^{j+1} c^{j-1} e_{k-j}(S) (\delta S)^\eta{}_\eta. \quad (\text{A.23b})$$

Thus we get,

$$\sum_{k=0}^n (-1)^k \delta e_k(\bar{S}) (\bar{S}^{n-k})^\rho{}_\nu = \sum_{k=0}^n (-1)^k \delta e_k(\bar{S}) c^{n-k} \delta^\rho{}_\nu, \quad (\text{A.24a})$$

$$= \sum_{j=1}^n \sum_{k=j}^n (-1)^k (-1)^{j+1} c^{n-(k-j)-1} e_{k-j}(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu. \quad (\text{A.24b})$$

Where in the second line we changed the order of the summation. Letting $l = k - j$, we can rewrite the sum as

$$\sum_{j=1}^n \sum_{k=j}^n (-1)^k (-1)^{j+1} c^{n-(k-j)-1} e_{k-j}(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu = \sum_{j=1}^n \sum_{l=0}^{n-j} (-1)^{l+1} c^{n-l-1} e_l(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu. \quad (\text{A.25})$$

Finally, changing the summation index one last time $d = n - j$, we get

$$\sum_{j=1}^n \sum_{l=0}^{n-j} (-1)^{l+1} c^{n-l-1} e_l(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu = \sum_{d=0}^{n-1} \sum_{l=0}^d (-1)^{l+1} c^{n-l-1} e_l(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu, \quad (\text{A.26a})$$

$$= \sum_{l=0}^{n-1} \sum_{d=l}^{n-1} (-1)^{l+1} c^{n-l-1} e_l(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu, \quad (\text{A.26b})$$

$$= \sum_{l=0}^{n-1} (-1)^{l+1} (n-l) c^{n-l-1} e_l(\bar{S}) (\delta S)^\eta{}_\eta \delta^\rho{}_\nu. \quad (\text{A.26c})$$

Using equation A.22 and A.26, the variation of the matrix $(Y_n)^\rho{}_\sigma(S)$ is expressed by

$$\delta(Y_n)^\rho{}_\sigma(S) = \sum_{l=0}^{n-1} (-1)^{l+1} (n-l) c^{n-l-1} e_l(\bar{S}) (\delta S^\eta{}_\eta \delta^\rho{}_\nu - \delta S^\rho{}_\nu). \quad (\text{A.27})$$

Let $N' = \sum_{l=0}^{n-1} (-1)^{l+1} (n-l) c^{n-l-1} e_l(\bar{S})$, we get

$$\delta(Y_n)^\rho{}_\sigma(S) = N' (\delta S^\eta{}_\eta \delta^\rho{}_\nu - \delta S^\rho{}_\nu). \quad (\text{A.28})$$

Given that the cosmological constant appearing in the background equation is equal to

$$\Lambda_g \bar{g}_{\mu\nu} = \frac{m^4}{m_g^2} \bar{g}_{\mu\rho} \sum_{n=0}^3 (-1)^n \beta_n (Y_n)^\rho{}_\nu(\bar{S}). \quad (\text{A.29})$$

The variation of the potential leads to

$$\frac{m^4}{m_g^2} g_{\mu\rho} \sum_{n=0}^3 (-1)^n \beta_n (Y_n)^\rho{}_\nu(S) = \Lambda_g \bar{g}_{\mu\nu} + \Lambda_g \delta g_{\mu\nu} + \frac{m^4}{m_g^2} N' \bar{g}_{\mu\rho} (\delta S^\eta{}_\eta \delta^\rho{}_\nu - \delta S^\rho{}_\nu). \quad (\text{A.30})$$

Letting $N = \frac{m^4}{m_g^2} N'$, we get

$$\frac{m^4}{m_g^2} V_{\mu\nu}^g(g, f, \beta_n) = \Lambda_g \bar{g}_{\mu\nu} + \Lambda_g \delta g_{\mu\nu} - N \bar{g}_{\mu\rho} (\delta S^\rho{}_\nu - \delta S^\eta{}_\eta \delta^\rho{}_\nu). \quad (\text{A.31})$$

A.3 Variation of the field equation

With all now in place we can prove the linearization of the field equation

$$G_{\mu\nu} + \frac{m^4}{m_g^2} V_{\mu\nu}^g(g, f, \beta_n) = 0, \quad (\text{A.32a})$$

$$\bar{G}_{\mu\nu} + \Lambda_g \bar{g}_{\mu\nu} + \delta G_{\mu\nu} + \Lambda_g \delta g_{\mu\nu} - N \bar{g}_{\mu\rho} (\delta S^\rho{}_\nu - \delta S^\eta{}_\eta \delta^\rho{}_\nu) = 0, \quad (\text{A.32b})$$

$$\delta G_{\mu\nu} + \Lambda_g \delta g_{\mu\nu} - N \bar{g}_{\mu\rho} (\delta S^\rho{}_\nu - \delta S^\eta{}_\eta \delta^\rho{}_\nu) = 0. \quad (\text{A.32c})$$

$$(\text{A.32d})$$

Where in the last equality is obtained by applying the background equation. Using the expression for the variation of the Einstein tensor A.15, we finally get

$$\varepsilon_{\mu\nu}{}^{\rho\sigma} \delta g_{\rho\sigma} - \Lambda_g (\delta g_{\mu\nu} - \frac{1}{2} \bar{g}_{\mu\nu} \bar{g}^{\rho\sigma} \delta g_{\rho\sigma}) - N \bar{g}_{\mu\rho} (\delta S^\rho{}_\nu - \delta S^\eta{}_\eta \delta^\rho{}_\nu) = 0. \quad (\text{A.33})$$

Given that

$$\bar{G}_{\mu\nu} = (1 + \alpha^2 c^2) \bar{g}_{\mu\nu}. \quad (\text{A.34})$$

we get

$$\bar{g}^{\mu\nu} = (1 + \alpha^2 c^2) \bar{G}^{\mu\nu}. \quad (\text{A.35})$$

and hence

$$\varepsilon_{\mu\nu}{}^{\rho\sigma} = (1 + \alpha^2 c^2) \bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma}. \quad (\text{A.36})$$

Where $\bar{\varepsilon}_{\mu\nu}{}^{\rho\sigma}$ is the differential operator defined with respect the metric $\bar{G}_{\mu\nu}$ in 2.21. Redefining the value of the parameter N to $\frac{N}{(1+\alpha^2 c^2)^2}$. we the first equation in 2.18. The second equation is obtained by by using the proportionality relation between the background metrics and the relation obeyed by the potentials, namely

$$V_{\mu\nu}^f(g, f, \beta_n) = V_{\mu\nu}^g(f, g, \beta_{4-n}). \quad (\text{A.37})$$

Appendix B

Proof of the Consistency of the Massless Mode Calculations With Shapiro Time Delay

We propose to prove that the contribution to time delay from the massless mode reproduces the Shapiro time delay.

$$\Delta t_G = \frac{\sqrt{s}}{16m_g^2\pi^2} \int_0^{2\pi} \int_0^\Lambda dq d\theta \frac{1}{q} e^{ibq \cos(\theta)}. \quad (\text{B.1})$$

In [22] this integral to calculated using dimensional perturbation and Schwinger perturbation. We propose another method to calculate this integral based on first kind Bessel function. using the identity

$$\int_0^{2\pi} d\theta e^{ibq \cos(\theta)} = 2\pi J_0(bq). \quad (\text{B.2})$$

we can rewrite the time delay as

$$\Delta t_G = \frac{\sqrt{s}}{8m_g^2\pi} \int_0^\Lambda dq \frac{J_0(bq)}{q}, \quad (\text{B.3a})$$

$$= \frac{\sqrt{s}}{16m_g^2\pi} \int_0^\Lambda dq \frac{J_0(bq) - 1}{q} + \frac{\sqrt{s}}{8m_g^2\pi} \int_0^\Lambda dq \frac{1}{q}. \quad (\text{B.3b})$$

The second part of the last equation is infrared divergent. This is expected as it reflects the long range behavior of the gravitational mode. Thus, a more suitable observable is the variation of time delay with respect to the impact parameter.

Using the asymptotic behavior of Bessel function

$$\int_0^\Lambda dq \frac{J_0(bq) - 1}{q} \xrightarrow{b\Lambda \rightarrow +\infty} -\ln\left(\frac{b\Lambda}{2}\right) - \gamma. \quad (\text{B.4})$$

Where γ is Euler-Mascheroni constant. using $m_g^2 = \frac{1}{16\pi G}$, we can rewrite the time delay as

$$\Delta t_G \approx 2G\sqrt{s} \ln\left(\frac{2}{b\Lambda}\right) + C + IR. \quad (\text{B.5a})$$

Where C is a constant depending on the center of mass energy s and IR is the infrared divergence part. Focusing on the b depended part we get

$$\Delta t_G \approx 2G\sqrt{s} \ln\left(\frac{2}{b\Lambda}\right), \quad (\text{B.6a})$$

$$= GM_s \ln\left(\frac{\mu^2}{b^2}\right). \quad (\text{B.6b})$$

Which is correct logarithmic dependence as in general relativity's Shapiro time delay. As a reminder Shapiro time delay is given by

$$\Delta t_{Shapiro} = 2GM_s \ln \frac{4r_E r_S}{b^2}. \quad (\text{B.7})$$

where r_E and r_S , are the distance of the emitter and receiver from the source. Imposing a correspondence between the massless contribution and the the classical time delay leads to imposing $\mu = 2\sqrt{r_E r_S}$.

There seems to be a factor of two missing, however as discussed more in details in [22]. In the classical derivation of the Shapiro time delay, one neglects surface terms in the integral, that become important in the case we are studying. Including them recover the agreement between the eikonal and classical calculations.

Appendix C

Time Delay in Quantum mechanics

C.1 Smith time delay calculation

In this section we retrace the derivation of the time delay in the review article by C.A.A. de Carvalho, H.M. Nussenzveig [32]. As mentioned before Smith proposed a definition of time delay as the difference in time a particle spend inside a region of radius R with and without interaction. In his original work, Smith only considered stationary solutions of the Schrodinger equation. First, the dwell time in a spherical region of radius R of the scattering center is defined as the ratio of the probability that a state is localized in this region to the inward probability flux into the region.

let $|\psi\rangle$ be the state describing the system and j the probability density vector, then the dwell time is given by:

$$t_R = \frac{\int_{|x|<R} |\psi|^2 d^3r}{\Phi_R}. \quad (\text{C.1})$$

with Φ_R , is the probability flux through the surface of the spherical region:

$$\Phi_R = - \oint j_{in} \cdot ds = - \frac{\hbar R^2}{2im} \oint \psi_{in} * \frac{\partial_{in} \psi}{\partial r} - \psi_{in} \frac{\partial \psi_{in}^*}{\partial r} d\Omega. \quad (\text{C.2})$$

In potential scattering theory, we usually consider asymptotic solutions to the Schrodinger equations. In its current form the nominator of this expression presents a problem: it indicates the necessity of integrating over the volume of the spherical region. This means that an expression for the wave function for arbitrary small distance r is needed. This difficulty could be surmounted by using Gauss theorem and changing the volume integral to a surface integral. Then, calculating the dwell time for sufficiently large radius R allows us to use the asymptotic expression for the wave function. To this end

consider the stationary Schrodinger equation:

$$H\psi = E\psi \quad (\text{C.3})$$

Taking the derivative with respect to energy we get,

$$H \frac{\partial \psi}{\partial E} = \psi + E \frac{\partial \psi}{\partial E}. \quad (\text{C.4})$$

So that:

$$\psi^* \psi = \psi^* H \psi - E \psi^* \frac{\partial \psi}{\partial E}, \quad (\text{C.5a})$$

$$\psi^* \psi = \psi^* H \psi - (H \psi)^* \frac{\partial \psi}{\partial E}, \quad (\text{C.5b})$$

$$\psi^* \psi = \frac{-\hbar^2}{2m} \left(\psi^* \frac{\partial}{\partial E} \nabla^2 \psi - \nabla^2 \psi^* \frac{\partial \psi}{\partial E} + \nabla \psi^* \frac{\partial}{\partial E} \nabla \psi - \nabla \psi^* \frac{\partial}{\partial E} \nabla \psi \right), \quad (\text{C.5c})$$

$$\psi^* \psi = \frac{-\hbar^2}{2m} \nabla \cdot \left(\psi^* \frac{\partial}{\partial E} \nabla \psi - \nabla \psi^* \frac{\partial \psi}{\partial E} \right). \quad (\text{C.5d})$$

Using Gauss theorem we get:

$$\int_{|x| < R} |\psi|^2 d^3r = \frac{\hbar^2 R^2}{2m} \oint \left(\frac{\partial \psi}{\partial E} \frac{\partial \psi}{\partial r} - \psi^* \frac{\partial^2 \psi}{\partial r \partial E} \right)_{r=R} d\Omega. \quad (\text{C.6})$$

Using the stationary solution to the free Schrodinger equation in spherical coordinates:

$$\psi_{in}(r, E) = \frac{e^{-ikr}}{r} A_l(\theta, \phi). \quad (\text{C.7})$$

with

$$k = \sqrt{\frac{2mE}{\hbar^2}} \quad (\text{C.8})$$

We can evaluate the flux of probability to the spherical region surrounding the scattering center,

$$\Phi_R = v B_E \quad (\text{C.9})$$

with:

$$v = \frac{\hbar k}{m}, \quad (\text{C.10a})$$

$$B_E = \int |A_l(\theta, \Phi)|^2 d\Omega. \quad (\text{C.10b})$$

Consider next the stationary Schrodinger equation with the potential V :

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi. \quad (\text{C.11})$$

If we assume that the potential decays quickly enough, then asymptotically we retrieve the free Schrodinger equation and thus the effect of the potential is restricted to change in the phase of the asymptotic wave function. Thus for a given value of the wave's angular momentum we get:

$$\psi(r, \theta, \phi) = 2i(-1)^l \frac{A_l(\theta, \phi)}{r} e^{i\eta_l(k)} \sin\left(kr + \eta_l(k) + \frac{\pi l}{2}\right). \quad (\text{C.12})$$

We therefore can calculate the probability of at a time t the state be localized inside the sphere of radius R around the scattering center,

$$\int_{|r|<R} |\psi|^2 d^3r = 2B_l(R + \frac{d\eta_l}{dk} - \frac{(-1)^l}{2k} \sin(2kr + 2\eta_l(k)) + O(\frac{1}{R})). \quad (\text{C.13})$$

Applying now the definition of the dwell time C.1 for state with angular momentum l ,

$$t_{R,l} = \frac{2R}{v} + \frac{2}{v} \frac{d\eta_l}{dk} - \frac{(-1)^l}{kv} \sin(2(kr + \eta_l)) + O(\frac{1}{R}). \quad (\text{C.14})$$

The last term in the expression of the dwell time of an interacting states is oscillatory and it links the uncertainty principle [36]. if we take the average of the dwell time given by:

$$\langle t_{R,l} \rangle = \frac{1}{R} \int_R^{2R} t_{R,l} dR. \quad (\text{C.15})$$

Then, the oscillatory term vanishes.

For a free propagating particle (in absence of the scattering potential) a similar calculation leads:

$$\langle t_{R,l}^0 \rangle = \frac{2R}{v}. \quad (\text{C.16})$$

Which is in agreement with the classical time it takes an unperturbed object to traverse a sphere of radius R .

Finally Smith defines the time delay a state experience due to the presence of a scattering potential as:

$$\Delta t_l = \lim_{R \rightarrow +\infty} [\langle t_{R,l} \rangle - \langle t_{R,l}^0 \rangle]. \quad (\text{C.17})$$

. In this case we get:

$$\Delta t_l = 2\hbar \frac{d\eta_l}{dk}. \quad (\text{C.18})$$

This indeed the same as Eisenbud-Wigner partial wave time delay.

Next we try to compute the derivative of the phase shift. Let ψ be the wave function of the system. If we note:

$$\psi_l(r, \theta, \phi) = A_l(\theta, \phi) \frac{u_l(E, r)}{r}. \quad (\text{C.19})$$

the $u_l(r)$ satisfies the following equation:

$$-\frac{\partial^2 u_l}{\partial r^2} + \left[\frac{l(l+1)}{r^2} + \frac{2m}{\hbar} V(r) \right] u_l = k^2 u_l. \quad (\text{C.20})$$

Consider next solution that satisfies

$$u_l(E, 0) = 0, \quad (\text{C.21a})$$

$$u_l(E, r) \rightarrow w_l(E, r) = \sin\left(kr + \frac{\eta_l}{2} - \frac{l\pi}{2}\right) (r \rightarrow \infty). \quad (\text{C.21b})$$

Consider next two solution for the radial equation for 2 different energies $u_l(E_1, r)$ and $u_l(E_2, r)$ which we note respectively as $u_{l,1}$ and $u_{l,2}$. Then we get

$$\frac{\partial}{\partial r} \left(u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2} \right) = u_{l,1} \frac{\partial^2 u_{l,2}}{\partial r^2} - \frac{\partial^2 u_{l,1}}{\partial r^2} u_{l,2}, \quad (\text{C.22a})$$

$$\frac{\partial}{\partial r} \left(u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2} \right) = (k_1^2 - k_2^2) u_{l,1} u_{l,2}. \quad (\text{C.22b})$$

where in going from C.22a to C.22b we used equation C.20. The same calculations for the corresponding $w_{l,1}$ and $w_{l,2}$ leads to

$$\frac{\partial}{\partial r} \left(w_{l,1} \frac{\partial w_{l,2}}{\partial r} - \frac{\partial w_{l,1}}{\partial r} w_{l,2} \right) = (k_1^2 - k_2^2) w_{l,1} w_{l,2}. \quad (\text{C.23})$$

Next consider:

$$J = \int_0^{+\infty} (u_{l,1} u_{l,2} - w_{l,1} w_{l,2}) dr. \quad (\text{C.24})$$

Using the 2 previous expression we get

$$J = \left[\frac{u_{l,1} \frac{\partial u_{l,2}}{\partial r} - \frac{\partial u_{l,1}}{\partial r} u_{l,2}}{k_1^2 - k_2^2} - \frac{w_{l,1} \frac{\partial w_{l,2}}{\partial r} - \frac{\partial w_{l,1}}{\partial r} w_{l,2}}{k_1^2 - k_2^2} \right]_0^{+\infty}, \quad (\text{C.25a})$$

$$J = w_{l,1(0)} \frac{\partial w_{l,2}}{\partial r}(0) - \frac{\partial w_{l,1}}{\partial r}(0) w_{l,2}(0). \quad (\text{C.25b})$$

Where in going from C.25a to C.25b the boundary conditions C.21 have been used. The first term in J is

$$J_1 = \frac{k_2 \cos\left(\eta_l(k_2) - \frac{\pi l}{2}\right) \sin\left(\eta_l(k_1) - \frac{\pi l}{2}\right)}{k_1^2 - k_2^2}. \quad (\text{C.26})$$

Taking the limit as k_1 approaches k_2 using the l'Hopital rule we get:

$$\lim_{k_1 \rightarrow k_2} \frac{\sin\left(\eta_l(k_1) - \frac{\pi l}{2}\right)}{k_1^2 - k_2^2} = \lim_{k_1 \rightarrow k_2} \frac{\frac{d\eta_l}{dk} \cos\left(\eta_l(k_1) - \frac{\pi l}{2}\right)}{2k_1}, \quad (\text{C.27a})$$

$$\lim_{k_1 \rightarrow k_2} \frac{\sin\left(\eta_l(k_1) - \frac{\pi l}{2}\right)}{k_1^2 - k_2^2} = \frac{1}{2k_2} \frac{d\eta_l}{dk}(k_2) \cos\left(\eta_l(k_2) - \frac{\pi l}{2}\right). \quad (\text{C.27b})$$

Thus we get

$$\lim_{k_1 \rightarrow k_2} J_1 = \frac{1}{2} \frac{d\eta_l}{dk}(k_2) \cos\left(\eta_l(k_2) - \frac{\pi l}{2}\right)^2. \quad (\text{C.28})$$

A similar calculation for the second term in the expression of J leads to

$$\lim_{k_1 \rightarrow k_2} J_2 = \lim_{k_1 \rightarrow k_2} \frac{k_1 \cos\left(\eta_l(k_1) - \frac{\pi l}{2}\right) \sin\left(\eta_l(k_2) - \frac{\pi l}{2}\right)}{k_1^2 - k_2^2}, \quad (\text{C.29a})$$

$$\lim_{k_1 \rightarrow k_2} J_2 = \frac{(-1)^l}{4k_2} \sin(2\eta_l(k_2)) - \frac{1}{2} \frac{d\eta_l}{dk}(k_2) \sin\left(\eta_l(k_2) - \frac{\pi l}{2}\right)^2. \quad (\text{C.29b})$$

Finally

$$\lim_{k_1 \rightarrow k_2} J = \lim_{k_1 \rightarrow k_2} J_1 - \lim_{k_1 \rightarrow k_2} J_2, \quad (\text{C.30a})$$

$$\lim_{k_1 \rightarrow k_2} J = \frac{1}{2} \frac{d\eta_l}{dk}(k_2) - \frac{(-1)^l}{4k_2} \sin(2\eta_l(k_2)). \quad (\text{C.30b})$$

Going back to the expression of J given by equation C.21, and taking the limit as k_1 approaches k_2 we get:

$$\lim_{k_1 \rightarrow k_2} J = \int_0^r (u_{l,2}^2 - w_{l,2}^2) dr. \quad (\text{C.31})$$

Thus we get:

$$\frac{d\eta_l}{dk} = (-1)^l \frac{\sin(2\eta_l(k))}{2k} + 2 \int_0^{+\infty} (u_l(E, r)^2 - w_l(E, r)^2) dr. \quad (\text{C.32})$$

Hence the time delay is given by

$$\Delta t_l = (-1)^l \frac{\sin(2\eta_l(k)) \hbar}{k} + 4\hbar \int_0^{+\infty} (u_l(E, r)^2 - w_l(E, r)^2) dr. \quad (\text{C.33})$$

C.2 The Time Delay Operator

The aim of this section is to give some insights on the derivation of the Wigner Smith operator, this section is again based on the review article [32] and more rigorous prove of the form of the Wigner Smith operator can be found in [37].

Consider a scattering system characterized by its free propagation Hamiltonian H_0 and the interaction Hamiltonian H . Next we define the free and interacting time evolution operators:

$$U_0(t) = \exp(-iH_0t), \quad (\text{C.34a})$$

$$U(t) = \exp(-iHt), \quad (\text{C.34b})$$

$$(\text{C.34c})$$

We can define the MØller operators:

$$\Omega_{\pm} = \lim_{t \rightarrow \mp\infty} U^{\dagger}(t)U_0(t). \quad (\text{C.35})$$

The MØller operator effects could be understood as following: When studying an interacting system of particles it's usually assumed that there excised a moment in the past where the different parts of the system where sufficiently far from each other that no interaction took place. Considering that in this moment no interaction took place the state of the system is described in terms of the non interacting Hamiltonian, in other words if the system at time t is described by a state $|\psi(t)\rangle$,

$$\lim_{t \rightarrow -\infty} |\psi(t)\rangle = |\phi_i\rangle. \quad (\text{C.36})$$

Similarly, if one is studying a scattering process, than the state of the system in the far future (away from the interaction region) is again described by the non interacting Hamiltonian. Thus, we get

$$\lim_{t \rightarrow +\infty} |\psi(t)\rangle = |\phi_f\rangle. \quad (\text{C.37})$$

If then suppose that state space of the interacting system, could be decomposed in 2 sets of complete eigen states: incoming and outgoing, The MØllerr operator Ω_- then link an eigenstate of the non-interacting system to the corresponding outgoing eigenstate of the interacting system. Similarly, the operator Ω_+ links an eigenstate of the non-interacting system to the corresponding incoming eigenstate of the interacting one.

This becomes clearer, if we use the following expressions:

$$\Omega_{\pm} = \sum |\psi_{\pm}^{\alpha}\rangle \langle \phi^{\alpha}|. \quad (\text{C.38})$$

where $|\psi_{\pm}^{\alpha}\rangle$ are the incoming(+) and outgoing (−) complete sets of eigen states of the interacting systems, and $|\phi^{\alpha}\rangle$ are the eigen states of non interacting systems it is not guaranteed that for a given scattering system the MØller operators exist. it is quite possible that the limit is equation It's important to note that the limit in the definition C.35 does not exist.

Let $|\psi(t)\rangle$ is state of the system as a time t , and let Σ be a region of space constrained within a radius R of the scattering center. let $\Phi_R(t)$ be the flux of probability to this region. We normalize the state such that:

$$\lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \Phi_R(t) dt = 1. \quad (\text{C.39})$$

Next consider P_{Σ} the projection operator to the region Σ :

$$P_{\Sigma} = \int_{|x| < R} |x\rangle \langle x| d^3x. \quad (\text{C.40})$$

then, the probability at time t to find the state in the region Σ is given by

$$p(t) = \langle \psi(t) | P_{\Sigma} | \psi(t) \rangle. \quad (\text{C.41})$$

The state $\psi(t)$ verifies:

$$|\psi(t)\rangle = U(t) |\psi(0)\rangle, \quad (\text{C.42a})$$

$$|\psi(0)\rangle = \Omega_{-} |\phi\rangle, \quad (\text{C.42b})$$

$$|\psi(t)\rangle = U(t) \Omega_{-} |\phi\rangle. \quad (\text{C.42c})$$

This follows because the state $|\psi(t)\rangle$ could be thought of at the time evolution of an incoming state $|\psi(0)\rangle$ which according to our previous remarks is deduced from the states of the free Hamiltonian via the MØller operator. Using this result we get

$$p(t) = \langle \phi | \Omega_{-}^{\dagger} U^{\dagger}(t) P_{\Sigma} U(t) \Omega_{-} | \phi \rangle. \quad (\text{C.43})$$

Using the Intertwining relation:

$$\Omega_{\pm} U = U_0 \Omega_{\pm}. \quad (\text{C.44})$$

We get

$$p(t) = \langle \phi | U_0^\dagger(t) \Omega_-^\dagger P_\Sigma \Omega_- U_0(t) | \Phi \rangle. \quad (\text{C.45})$$

The dwell time is then defined as the integral over time of the $p(t)$

$$T_R = \int_{-\infty}^{+\infty} p(t) dt = \int_{-\infty}^{+\infty} \langle \phi | U_0^\dagger(t) \Omega_-^\dagger P_\Sigma \Omega_- U_0(t) | \Phi \rangle dt. \quad (\text{C.46})$$

For the free propagating particle we get:

$$T_{0R} = \int_{-\infty}^{+\infty} \langle \phi | U_0^\dagger(t) P_\Sigma U_0(t) | \Phi \rangle dt. \quad (\text{C.47})$$

and the time delay is therefore given as:

$$\tau = \lim_{R \rightarrow +\infty} T_R - T_{R0} = \lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \langle \phi | U_0^\dagger(t) (\Omega_-^\dagger P_\Sigma \Omega_- - P_\Sigma) U_0(t) | \Phi \rangle dt. \quad (\text{C.48})$$

This leads to the definition of the time delay operator by its expectation values for given states

$$\langle \psi | Q | \psi \rangle = \lim_{R \rightarrow +\infty} \int_{-\infty}^{+\infty} \langle \phi | U_0^\dagger(t) (\Omega_-^\dagger P_\Sigma \Omega_- - P_\Sigma) U_0(t) | \Phi \rangle dt. \quad (\text{C.49})$$

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